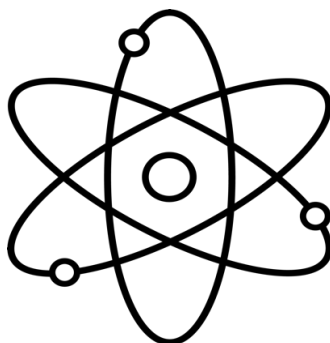




SCIENCE

Invitational A • 2015



GENERAL DIRECTIONS:

- DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
- Ninety minutes should be ample time to complete this contest, but since it is not a race, contestants may take up to two hours. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet.
- You may place as many notations as you desire anywhere on the test paper except on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- On the back of this page is printed a copy of the periodic table of the elements. You may wish to refer to this table in answering the questions, and if needed, you may use the atomic weights and atomic numbers from the table. Other scientific relationships are listed also.
- Silent hand-held calculators that do not need external wall plugs may be used. Graphing calculators that do not have built-in or stored functionality that provides additional scientific information are allowed. Small hand-held computers are not permitted. Calculators that accept memory cards or memory sticks are not permitted. Each contestant may bring one spare calculator. All memory must be cleared.

SCORING:

All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for an incorrect answer.

Periodic Table of the Elements

1A 1		2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.008																		2 He 4.003
3 Li 6.941	4 Be 9.012												5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12		13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41		31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41		49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59		81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (285)							

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Some Standard Properties of Water

property	symbol	value
density of water	ρ_{water}	1.000 g cm ⁻³
density of ice	ρ_{ice}	0.9167 g cm ⁻³
specific heats		
ice	C_{ice}	2.09 J g ⁻¹ K ⁻¹
water	C_{water}	4.184 J g ⁻¹ K ⁻¹
steam	C_{steam}	2.03 J g ⁻¹ K ⁻¹
heat of fusion	ΔH_{fus} or L_f	334 J g ⁻¹
heat of vaporization	ΔH_{vap} or L_v	2260 J g ⁻¹
index of refraction	n	1.33
autoionization	K_w	1.0×10^{-14}

Pressure		
1 atm	=	760 torr
	=	101325 Pa
	=	14.7 psi
1 bar	=	10 ⁵ Pa
	=	100 kPa

Energy		
1 cal	=	4.184 J
1 L atm	=	101.325 J
1 Cal	=	1 kcal
1 hp	=	746 W
1 eV	=	1.602×10^{-19} J

Various Physical Constants

property	symbol	value
universal gas constant	R	8.314 J mol ⁻¹ K ⁻¹ 62.36 L torr mol ⁻¹ K ⁻¹ 0.08206 L atm mol ⁻¹ K ⁻¹ 1.987 cal mol ⁻¹ K ⁻¹
Planck's constant	h	6.626×10^{-34} J s 4.136×10^{-15} eV s
Planck's reduced constant	$h/2\pi$	1.054×10^{-34} J s 6.582×10^{-16} eV s
Boltzmann constant	k_B	1.38×10^{-23} J K ⁻¹
Stefan-Boltzmann	σ	5.67×10^{-8} W m ⁻² K ⁻⁴
speed of light	c	3.00×10^8 m s ⁻¹
speed of sound (at 20°C)	v_{air}	343 m s ⁻¹
acceleration of gravity	g	9.80 m s ⁻²
gravitational constant	G	6.67×10^{-11} N m ² kg ⁻²
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
elementary charge	e	1.602×10^{-19} C
Faraday	F	96485 C mol ⁻¹
Coulomb's law constant	k	8.988×10^9 N m ² C ⁻²
Rydberg constant	R	2.178×10^{-18} J

Some Other Conversion Factors

1 in	=	2.54 cm
1 lb	=	453.6 g
1 mi	=	5280 ft = 1.609 km
1 gal	=	4 quarts = 231 in ³ = 3.785 L

property	symbol	value
electron rest mass	m_e	9.11×10^{-31} kg 0.000549 u 0.511 MeV c ⁻²
proton mass	m_p	1.6726×10^{-27} kg 1.00728 u 938.3 MeV c ⁻²
neutron mass	m_n	1.6749×10^{-27} kg 1.008665 u 939.6 MeV c ⁻²
atomic mass unit	u	1.6605×10^{-27} kg 931.5 MeV c ⁻²
earth mass		5.972×10^{24} kg
earth radius		6.371×10^6 m
moon mass		7.348×10^{22} kg
sun mass		1.989×10^{30} kg
distance earth-moon		3.844×10^8 m
distance earth-sun		1.496×10^{11} m
permittivity of free space	ϵ_0	8.85×10^{-12} F m ⁻¹
permeability of free space	μ_0	$4\pi \times 10^{-7}$ T m A ⁻¹

Some Average Bond Energies (kJ/mol)

C-H	413	C-C	346	C-Cl	339	C-N	305
O-H	463	C=C	602	C-Br	285	N=N	418
N-H	391	C≡C	835	O=O	498	H-H	436
C-O	358	C=O	799	C≡O	1072	Br-Br	193
H-Cl	432	S-H	347	N≡N	945	Cl-Cl	242
H-Br	366	H-I	299	C≡N	887	I-I	151

Biology Questions (1 – 20)

1. Isotopes of the same element have different mass numbers but identical atomic numbers because isotopes of the same element have a
A) different number of protons and neutrons.
B) different number of protons.
C) different structure of electrons residing in electron shells.
D) different number of electrons and protons.
E) different level of radioactivity.
2. Fatty acids are long hydrocarbon chains with a _____ functional group.
A) hydroxyl
B) carboxyl
C) carbonyl
D) sulfhydryl
E) amino
3. A molecule made of covalent bonds in which the sharing of electrons is equal is
A) polar
B) aqueous
C) hydrophilic
D) hydrophobic
E) non polar
4. As the pH of a solution decreases, the hydroxyl ion concentration
A) increases
B) decreases
C) is not affected
5. The process by which particles move from areas of higher concentration to areas of lower concentration is called
A) osmosis
B) a concentration gradient
C) diffusion
D) mass movement
E) entropy
6. The closest relatives to the land plants are the _____
A) green algae
B) ferns
C) mosses
D) aquatic plants
E) seedless plants
7. Energy present in a system that is unable to perform work is called
A) entropy
B) heat
C) potential energy
D) equilibrium energy
E) radiant energy
8. The sum total of all genes in a population is known as
A) Hardy-Weinberg equilibrium
B) the allele pool
C) the gene pool
D) the biome
E) the genome
9. An autosome is a chromosome that
A) is found in all human cells
B) is found only in human gametes
C) is found only in females
D) is found only in males
E) determines the gender of humans
10. The three basic plant organs are the _____
A) cells, tissues, and meristems
B) roots, shoots, and flowers
C) roots, stems, and leaves
D) xylem, phloem, and roots
E) buds, nodes, and internodes

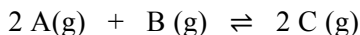
11. Which of the following is an organelle found in a prokaryotic cell?
A) ribosome
B) DNA
C) plasma membrane
D) All the above
E) None of the above
12. Nutrients that cannot be biosynthesized and must be consumed in the diet are known as the _____ nutrients.
A) limiting
B) factorial
C) vitaminal
D) essential
E) required
13. A circulatory system has three basic components: the circulatory fluid, the circulatory vessels, and the _____.
A) interstitial fluid
B) lungs
C) heart
D) blood
E) arteries and veins
14. Bacteria and Archaea are both considered
A) prokaryotes
B) halophytes
C) eubacteria
D) eukaryotes
E) autotrophic
15. The nerve cells that transfer information within the body are known as _____.
A) neurons
B) dendrites
C) ganglia
D) synapses
E) effectors
16. What molecules are produced during translation?
A) DNA
B) RNA
C) ribosomes
D) peptides
E) amino acids
17. The inability to interbreed, known as reproductive isolation, is a central tenet of the _____ species concept.
A) reproductive
B) biological
C) recognition
D) phylogenetic
E) genetic
18. In binomial nomenclature, the first of the two names is the _____ and should be _____.
A) specific epithet, capitalized
B) species; underlined
C) species; capitalized
D) genus; capitalized
E) genus; lower case
19. How many phosphate groups are found in a single nucleotide?
A) 0 in DNA, 0 in RNA
B) 1 in DNA, 1 in RNA
C) 1 in DNA, 2 in RNA
D) 2 in DNA, 1 in RNA
E) 2 in DNA, 2 in RNA
20. The field of biology that identifies and classifies organisms is known as
A) phylogenetics
B) nomenclology
C) systems
D) biogeography
E) taxonomy

Chemistry Questions (21 – 40)

21. What is the name for the compound CoSO_3 ?
- copper(II) sulfate
 - cobalt(II) sulfite
 - chromium(I) sulfite
 - cobalt(II) sulfate
 - copper(II) sulfite
22. What is the chemical formula for ammonium iodate?
- Am_2I
 - NH_3I
 - NH_4IO_4
 - NH_4IO_3
 - $(\text{NH}_4)_2\text{IO}_3$
23. What is the molar mass of $\text{Fe}(\text{CH}_3\text{CO}_2)_3$?
- 233.0 g/mol
 - 114.9 g/mol
 - 173.9 g/mol
 - 344.7 g/mol
 - 185.0 g/mol
24. What is the percentage of carbon by mass in caffeine ($\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$)?
- 6.18%
 - 51.0%
 - 49.5%
 - 33.3%
 - 56.2%
25. Which of the following atoms will have the largest atomic radius?
- Li
 - Zn
 - Br
 - Mg
 - Ca
26. Balance the following reaction with whole numbers as coefficients.
- $$\text{C}_4\text{H}_{10}(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$$
- What are the respective coefficients?
- 2 : 13 : 8 : 10
 - 1 : 6 : 4 : 5
 - 1 : 9 : 4 : 5
 - 2 : 14 : 7 : 10
 - 3 : 19 : 12 : 15
27. Refer to the reaction shown in the previous question. What type of reaction is this?
- neutralization
 - precipitation
 - combustion
 - metathesis
 - polymerization
28. Balance the following equation.
- $$\text{BF}_3 + \text{H}_2\text{O} \rightarrow \text{B}_2\text{O}_3 + \text{HF}$$
- What is the maximum number of grams of B_2O_3 that can be formed when 100 g of BF_3 and 50 g of H_2O completely react?
- 64.4 g
 - 51.3 g
 - 94.4 g
 - 48.5 g
 - 72.8 g
29. KNO_3 is a major component of fertilizer and can be made by the following reaction
- $$4\text{KCl} + 4\text{HNO}_3 + \text{O}_2 \rightarrow 4\text{KNO}_3 + 2\text{Cl}_2 + 2\text{H}_2\text{O}$$
- How many pounds of KCl are needed to make 500 lbs of KNO_3 ?
- 678 lbs
 - 413 lbs
 - 510 lbs
 - 2000 lbs
 - 369 lbs

30. What is the electron configuration of tin?
- $[\text{Ar}] 4s^2 3d^2$
 - $[\text{Ar}] 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^3$
 - $[\text{Ar}] 4s^2 3d^{10} 4p^6 5s^1 4d^9 5p^4$
 - $[\text{Ar}] 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^0 6s^2$
 - $[\text{Ar}] 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^2$
31. What is the energy of a photon with a wavelength of 440 nm?
- $5.15 \times 10^{-19} \text{ J}$
 - $6.82 \times 10^{14} \text{ J}$
 - $2.16 \times 10^{-18} \text{ J}$
 - $4.52 \times 10^{-19} \text{ J}$
 - $2.92 \times 10^{-22} \text{ J}$
32. What is the electron dot formula for nitrate?
- $\left[\begin{array}{c} \text{:}\ddot{\text{O}}\text{:} \\ \text{:}\ddot{\text{O}}\text{:N}\text{:}\ddot{\text{O}}\text{:} \end{array} \right]^-$
 - $\left[\begin{array}{c} \text{:}\ddot{\text{O}}\text{:} \\ \text{:}\ddot{\text{O}}\text{:N}\text{:}\ddot{\text{O}}\text{:} \end{array} \right]^-$
 - $\left[\begin{array}{c} \text{:}\ddot{\text{O}}\text{:} \\ \text{:}\ddot{\text{O}}\text{:N}\text{:}\ddot{\text{O}}\text{:} \end{array} \right]^-$
 - $\left[\begin{array}{c} \text{:}\ddot{\text{N}}\text{:} \\ \text{:}\ddot{\text{O}}\text{:O}\text{:}\ddot{\text{O}}\text{:} \end{array} \right]^-$
33. Which of the following molecules is not polar?
- NH_3
 - SF_4
 - BeF_2
 - PH_3
 - CH_2O
34. An exothermic reaction is at equilibrium. As the reaction is slowly heated, which of the following must be true about the reactants and products?
- the products will increase in concentration
 - the reactants will increase in concentration
 - the concentrations of both reactants and products will remain constant
35. 75 mL of hydrochloric acid is titrated to the end point with 21.92 mL of 0.0155 M NaOH. What is the concentration of the hydrochloric acid?
- 0.00453 M
 - 0.00227 M
 - 0.00906 M
 - 0.0530 M
 - 0.0155 M
36. A 5 liter gas container is filled with 10 g of ethane (C_2H_6) at 25°C . What is the pressure of the methane gas in this container?
- 0.137 atm
 - 14.7 atm
 - 8.27 atm
 - 3.91 atm
 - 1.63 atm
37. An evacuated container is filled with equal masses of argon and neon. What is the mole fraction of argon in this mixture?
- 41.1 %
 - 33.6 %
 - 50.0 %
 - 66.4 %
 - 58.9 %
38. What is the molar solubility of $\text{Pb}(\text{OH})_2$ if the value of K_{sp} is equal to 1.43×10^{-20} ?
- $1.20 \times 10^{-10} \text{ mol/L}$
 - $8.46 \times 10^{-11} \text{ mol/L}$
 - $1.53 \times 10^{-7} \text{ mol/L}$
 - $3.72 \times 10^{-8} \text{ mol/L}$
 - $2.43 \times 10^{-7} \text{ mol/L}$
39. What is the pH of a $2.9 \times 10^{-4} \text{ M}$ solution of KOH?
- 10.46
 - 10.54
 - 3.54
 - 10.76
 - 3.24

40. Which of the equations shown is equal to the equilibrium constant, K_p , for the following gas phase chemical reaction?



- A) $\frac{P_C}{P_A P_B}$ D) $\frac{P_C^2}{P_A^2 P_B}$
 B) $\frac{2P_A P_B}{2P_C}$ E) $\frac{P_A^2 P_B}{P_C^2}$
 C) $\frac{2P_C}{2P_A P_B}$

Physics Questions (41 – 60)

On this contest $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors in the $+x$, $+y$ and $+z$ directions respectively, and unless otherwise indicated, you may neglect air resistance in every problem.

41. According to Tyson, who discovered Pluto?
 A) Venetia Burney
 B) Percival Lowell
 C) Henry Madan
 D) Clyde Tombaugh
 E) Herbert Hall Turner
42. According to Tyson, which of the following discovered in 1789 is the heaviest atom in nature?
 A) lead
 B) mercury
 C) platinum
 D) plutonium
 E) uranium
43. According to Tyson, which scientist led a team of physicists to produce a new element in 1941?
 A) Philip H. Abelson
 B) Percival Lowell
 C) Edwin M. McMillan
 D) Glen T. Seaborg
 E) Herbert Hall Turner
44. According to Tyson, he stated that he believes that Americans love of Pluto is to be blamed on
 A) being discovered by an American astronomer
 B) being named after the roman god of the dead and underworld
 C) the popularity and efficacy of Pluto water
 D) the popularity of Mickey's dog in a cartoon by Walt Disney
 E) the use of plutonium in the first atomic bomb used in World War II
45. Pirates starting from their home port travel 75.0 m at 60.0° E of N then 57.3 m at 25.0° S of W and finally due south for 25.0 m. With the origin of the coordinate system at their home port and with E and N in the $+x$ and $+y$ directions respectively on their treasure map, where as measured in standard position will the pirates place the X on their treasure map?
 A) 17.5 m, 318°
 B) 36.7 m, 343°
 C) 86.7 m, 323°
 D) 123 m, 318°
 E) 146 m, 318°
46. If you throw a ball upward, at the top of the vertical motion the ball is momentarily motionless. Which of the following is true at this point?
 A) acceleration is zero
 B) acceleration is 9.8 m/s^2 downward
 C) acceleration is 9.8 m/s^2 upward
 D) velocity is constant
 E) velocity is non-zero
47. A roadrunner runs along a straight road from the coyote. While watching you notice that the roadrunner's coordinate position is described by $x(t) = 23.00 \text{ m} + (7.250 \text{ m/s}^2) t^2$, what is the roadrunner's displacement from 2.000 s to 3.000 s?
 A) 7.25 m
 B) 29.00 m
 C) 36.25 m
 D) 65.25 m
 E) 140.25 m

48. A man pulls a 650 N crate by pulling on a rope at an angle of 30.0° above the horizontal. The coefficient of kinetic friction between the crate and the horizontal floor is 0.370. How hard does the man have to pull to keep the crate moving with a constant velocity?
- A) 208 N
B) 229 N
C) 293 N
D) 325 N
E) 650 N
49. While crossing a river a boat heads due north at a speed of 40 km/hr relative to the water. If the river flows from west to east at 18 km/hr relative to the shore then what is the velocity of the boat relative to shore?
- A) 22 km/hr, 24° E of N
B) 44 km/hr, 24° E of N
C) 44 km/hr, 66° W of N
D) 58 km/hr, 24° W of N
E) 58 km/hr, 66° E of N
50. A 75.0 kg man rides a Ferris wheel with a diameter of 20.0 m and the wheel makes one revolution in 12.0 s. What is the apparent weight of the man at the top of the Ferris wheel? You may assume the seat remains upright during the ride.
- A) 206 N
B) 268 N
C) 529 N
D) 735 N
E) 941 N
51. A particle that moves in the xy -plane undergoes a displacement of $\Delta \mathbf{r} = (3.0 \mathbf{i} + 2.0 \mathbf{j})$ m while being acted upon by a constant force $\mathbf{F} = (4.0 \mathbf{i} + 4.0 \mathbf{j})$ N. What is the work done by this force during the displacement?
- A) 4.0 J
B) 20 J
C) $(7.0 \mathbf{i} + 6.0 \mathbf{j})$ J
D) $(12 \mathbf{i} - 8.0 \mathbf{j})$ J
E) $(12 \mathbf{i} + 8.0 \mathbf{j})$ J
52. A neutron travels at 2.6×10^7 m/s collides head on with a carbon nucleus that is initially at rest. Given that a neutron has an approximate mass of 1.0 u and a carbon nucleus has an approximate mass of 12 u. What is the magnitude of the velocity of the neutron after the collision if the collision is elastic and one-dimensional?
- A) 0.0 m/s
B) 4.0×10^6 m/s
C) 2.2×10^7 m/s
D) 3.0×10^7 m/s
E) 3.2×10^7 m/s
53. An old school record player rotates at 5600 rev/min. How much time does it take a record with a diameter of 25 cm to rotate through 180° ?
- A) 8.9×10^{-5} s
B) 2.7×10^{-3} s
C) 5.4×10^{-3} s
D) 1.1×10^{-2} s
E) 3.2×10^{-1} s
54. A weight W at the end of a string of length L swings from a ceiling hook. If T is the tension found in the string then what is the magnitude of the net torque about the hook when the string makes an angle ϕ with vertical?
- A) WL
B) $WL \cos \phi$
C) $WL/(T \cos \phi)$
D) $WL \sin \phi$
E) $WL/(T \sin \phi)$
55. A 6.00 m long cable is used to lift very heavy objects. When a crate of 500 kg is lifted then the cable has a cross sectional area of 0.250 cm^2 and stretches by 0.570 cm. What is Young's modulus for the cable?
- A) 4.85×10^{-12} Pa
B) 9.50×10^{-4} Pa
C) 1.29×10^2 Pa
D) 1.96×10^8 Pa
E) 2.06×10^{11} Pa

56. If a glass of water is left outside to freeze into ice thus lowering the entropy of the glass of water and if we only consider energy exchange between the glass of water and the surrounding air, then which of the following is true of this process
- The entropy of the surrounding air stays the same.
 - The entropy of both the glass of water and the surrounding air decrease.
 - The absolute value of the change in entropy of the glass of water is equal to the absolute value of the change in entropy of the surrounding air.
 - The absolute value of the change in entropy of the glass of water is more than the absolute value of the change in entropy of the surrounding air.
 - The absolute value of the change in entropy of the glass of water is less than the absolute value of the change in entropy of the surrounding air.
57. A point charge $q_1 = +4.00 \mu\text{C}$ is located on the positive y -axis at $y = 0.300 \text{ m}$ and a second charge $q_2 = +4.00 \mu\text{C}$ is located at the origin. Find the magnitude of the total force that q_1 and q_2 exert on a third charge $q_3 = +6.00 \mu\text{C}$ located on the x -axis at $x = 0.500 \text{ m}$.
- 0.326 N
 - 1.08 N
 - 1.32 N
 - 1.44 N
 - 1.50 N
58. Determine the rms speed of the following group of molecules. The molecules have speeds of 300 m/s, 400 m/s, 450 m/s, 500 m/s, 650 m/s, and 750 m/s.
- 450 m/s
 - 508 m/s
 - 530 m/s
 - 750 m/s
 - 281250 m/s
59. A copper wire has a diameter of 0.96 mm and a cross sectional area of $7.2 \times 10^{-7} \text{ m}^2$. If the wire has a resistance of 1.2Ω at 23°C then what is the resistance of the wire at 100°C ? The resistivity of copper is $1.7 \times 10^{-8} \Omega\cdot\text{m}$ at 23°C and the temperature coefficient of resistivity of copper is $0.0039/\text{K}$
- $4.7 \times 10^{-3} \Omega$
 - $3.6 \times 10^{-1} \Omega$
 - $5.8 \times 10^{-1} \Omega$
 - 1.6Ω
 - 1.8Ω
60. An electron moves at high (non-relativistic) speed through a magnetic field that is oriented perpendicular to its direction of travel. Which of the following will change?
- acceleration and velocity
 - acceleration and kinetic energy
 - charge and mass
 - kinetic energy and velocity
 - total energy

UIL HIGH SCHOOL SCIENCE CONTEST

Contestant Answer Sheet

Contestant # _____

9 10 11 12
CONTESTANT GRADE LEVEL

A 2A 3A 4A 5A
CONFERENCE

PLEASE WRITE ANSWERS WITH CAPITAL LETTERS

- | | | |
|-----------|-----------|-----------|
| 1. _____ | 21. _____ | 41. _____ |
| 2. _____ | 22. _____ | 42. _____ |
| 3. _____ | 23. _____ | 43. _____ |
| 4. _____ | 24. _____ | 44. _____ |
| 5. _____ | 25. _____ | 45. _____ |
| 6. _____ | 26. _____ | 46. _____ |
| 7. _____ | 27. _____ | 47. _____ |
| 8. _____ | 28. _____ | 48. _____ |
| 9. _____ | 29. _____ | 49. _____ |
| 10. _____ | 30. _____ | 50. _____ |
| 11. _____ | 31. _____ | 51. _____ |
| 12. _____ | 32. _____ | 52. _____ |
| 13. _____ | 33. _____ | 53. _____ |
| 14. _____ | 34. _____ | 54. _____ |
| 15. _____ | 35. _____ | 55. _____ |
| 16. _____ | 36. _____ | 56. _____ |
| 17. _____ | 37. _____ | 57. _____ |
| 18. _____ | 38. _____ | 58. _____ |
| 19. _____ | 39. _____ | 59. _____ |
| 20. _____ | 40. _____ | 60. _____ |

BIOLOGY SCORE

CHEMISTRY SCORE

PHYSICS SCORE

OVERALL SCORE:

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY**

INVITATIONAL A • 2015

- | | | |
|-------|-------|-------|
| 1. A | 21. B | 41. D |
| 2. B | 22. D | 42. E |
| 3. E | 23. A | 43. D |
| 4. B | 24. C | 44. D |
| 5. C | 25. E | 45. A |
| 6. A | 26. A | 46. B |
| 7. B | 27. C | 47. C |
| 8. C | 28. B | 48. B |
| 9. A | 29. E | 49. B |
| 10. C | 30. E | 50. C |
| 11. E | 31. D | 51. B |
| 12. D | 32. A | 52. C |
| 13. C | 33. C | 53. C |
| 14. A | 34. B | 54. D |
| 15. A | 35. A | 55. E |
| 16. D | 36. E | 56. E |
| 17. B | 37. B | 57. D |
| 18. D | 38. C | 58. C |
| 19. B | 39. A | 59. D |
| 20. E | 40. D | 60. A |

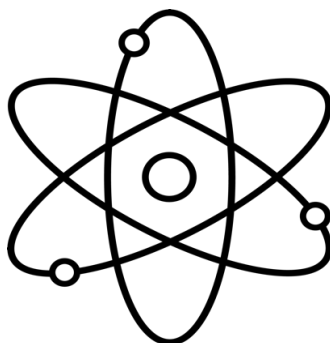
PHYSICS KEY for Science Contest • Invitational A • 2015

41. (D) “At about four in the afternoon on February 18, 1930, 24-year-old Clyde W. Tombaugh, a farm boy and amateur astronomer from Illinois, discovered on the sky what would be shortly named for the Roman god of the underworld.”
42. (E) “By 1789, just eight years after Herschel discovered Uranus, Martin Klaproth, of Germany, discovered the heaviest atom in nature. In need of a name, and with the planet Uranus fresh on people’s minds, the element uranium would ultimately land at slot 92 on the periodic table...”
43. (D) “Eleven years later, in 1941, a team of physicists led by Glen T. Seaborg manufactured a new element for the periodic table ... while working at the University of California at Berkeley’s cyclotron – one of the world’s preeminent atom smashers.”
44. (D) “Whether or not Walt Disney was thinking about the cosmos when he named his dog is not important here. What matters is that the seeds were sown for planet Pluto to receive a level of attention from the American public that far exceeds its astrophysical significance in the solar system.”
45. (A) Adding by components gives: $R_x = +75\sin(60^\circ) - 57.3\cos(25^\circ) = +13.02\text{m}$ &
 $R_y = +75\cos(60^\circ) - 57.3\sin(25^\circ) - 25 = -11.72\text{m} \Rightarrow R = [(13.02)^2 + (-11.72)^2]^{1/2} = 17.5\text{m}$ &
 $\theta = \tan^{-1}(-11.72/13.02) = -41.98^\circ + 360^\circ = 318^\circ$
46. (B) The acceleration throughout the entire motion is a constant 9.8m/s^2 down.
47. (C) $\Delta x = [23 + 7.25(3)^2] - [23 + 7.25(2)^2] = +36.25\text{ m}$
48. (B) From the FBD & N2L: (vertically) $+F_N + F_T \sin(30^\circ) - F_G = 0 \Rightarrow F_N = 650 - F_T \sin(30^\circ)$
 (horizontally) $+F_T \cos(30^\circ) - F_{\text{frk}} = 0$ & with $F_{\text{frk}} = \mu_k F_N = \mu_k [650 - F_T \sin(30^\circ)]$
 $\Rightarrow F_T = \mu_k(650)/[\cos(30^\circ) - \mu_k \sin(30^\circ)] = 229\text{ N}$
49. (B) By relative velocity addition: $v = [40^2 + 18^2]^{1/2} = 44\text{km/hr}$ at $\theta = \tan^{-1}(18/40) = 24^\circ$ E of N
50. (C) From the FBD & N2L: (radially) $+F_G - F_N = m(+a_c) \Rightarrow F_N = m(g - v^2/r)$
 where $v = 2\pi r/T = 2\pi(10)/(12) = 5.236\text{ m/s} \Rightarrow F_N = 75[9.8 - 5.236^2/10] = 529\text{ N}$
51. (B) $W = \mathbf{F} \cdot \Delta \mathbf{r} = (4)(3) + (4)(2) = 20\text{ J}$
52. (C) By conservation of momentum: $(1.0)(+2.6\text{E}+7) + 0 = (1.0)(+v_1') + (12)(+v_2')$ & for a 1-D elastic collision by the relative velocity equation: $(+2.6\text{E}+7) - (0) = (+v_2') - (+v_1')$
 $\Rightarrow v_1' = -11(+2.6\text{E}+7)/13 = -2.2\text{E}+7\text{ m/s}$
53. (C) Note: 180° is $\frac{1}{2}$ of a revolution and will take $\frac{1}{2}$ of the period to complete. So,
 $t = \frac{1}{2}T = \frac{1}{2}[(1\text{ min})/(5600\text{ rev})][(60\text{ s})/(1\text{ min})] = 5.4\text{E}-3\text{ s}$
54. (D) From the FBD the torque about the hook from the tension in the string is zero, thus the net torque is due to the gravitational force: $\Sigma \tau = \tau_G = \pm r F_G \sin \theta = \pm L W \sin \varphi$
55. (E) By $Y = \text{Stress/strain} = (F/A)/(\Delta L/L_0) = [(750)(9.8)(6)]/[(0.25\text{E}-4)(0.57\text{E}-2)] = 2.06\text{E}+11\text{ Pa}$
56. (E) By the second law of thermodynamics for an irreversible process the total entropy of an isolated system must increase, or $\Delta S \geq 0$. Thus the absolute value of the change in entropy of the glass of water must be less than the absolute value of the change in entropy of the surrounding air.
57. (D) The magnitudes of the electric forces are: $F_{23} = (8.99\text{E}+9)(4\text{E}-6)(6\text{E}-6)/(0.5)^2 = 0.8630\text{ N}$ &
 $F_{13} = (8.99\text{E}+9)(4\text{E}-6)(6\text{E}-6)/(0.3^2 + 0.5^2) = 0.6346\text{ N}$ from the diagram the angle for F_{13} is
 $\theta = \tan^{-1}(0.3/0.5) = 30.96^\circ$ and the net force is found by vector addition of these two force vectors:
 $\Sigma F_x = +0.8630 + 0.6346 \cos 30.96^\circ = 1.4072\text{ N}$ & $\Sigma F_y = +0 - 0.6346 \sin 30.96^\circ = -0.3265\text{ N}$
 $\Rightarrow \Sigma F = [(1.4072)^2 + (-0.3265)^2]^{1/2} = 1.44\text{ N}$
58. (C) By definition an rms average is the square root of the average of the squares of the values, thus
 $v_{\text{rms}} = [(300^2 + 400^2 + 450^2 + 500^2 + 650^2 + 750^2)/6]^{1/2} = 530\text{ m/s}$
59. (D) By $R = R_0[1 + \alpha \Delta T] = 1.2[1 + (0.0039)(100-23)] = 1.6\ \Omega$
60. (A) A charged particle moving perpendicularly through a magnetic field moves in uniform circular motion. Thus both the velocity and acceleration vectors will change directions during the motion.



SCIENCE

Invitational B • 2015



GENERAL DIRECTIONS:

- DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
- Ninety minutes should be ample time to complete this contest, but since it is not a race, contestants may take up to two hours. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet.
- You may place as many notations as you desire anywhere on the test paper except on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- On the back of this page is printed a copy of the periodic table of the elements. You may wish to refer to this table in answering the questions, and if needed, you may use the atomic weights and atomic numbers from the table. Other scientific relationships are listed also.
- Silent hand-held calculators that do not need external wall plugs may be used. Graphing calculators that do not have built-in or stored functionality that provides additional scientific information are allowed. Small hand-held computers are not permitted. Calculators that accept memory cards or memory sticks are not permitted. Each contestant may bring one spare calculator. All memory must be cleared.

SCORING:

All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for an incorrect answer.

Periodic Table of the Elements

1A 1		2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.008																		2 He 4.003
3 Li 6.941	4 Be 9.012												5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12		13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41		31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41		49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59		81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (285)							

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Some Standard Properties of Water

property	symbol	value
density of water	ρ_{water}	1.000 g cm ⁻³
density of ice	ρ_{ice}	0.9167 g cm ⁻³
specific heats		
ice	C_{ice}	2.09 J g ⁻¹ K ⁻¹
water	C_{water}	4.184 J g ⁻¹ K ⁻¹
steam	C_{steam}	2.03 J g ⁻¹ K ⁻¹
heat of fusion	ΔH_{fus} or L_f	334 J g ⁻¹
heat of vaporization	ΔH_{vap} or L_v	2260 J g ⁻¹
index of refraction	n	1.33
autoionization	K_w	1.0×10^{-14}

Pressure		
1 atm	=	760 torr
	=	101325 Pa
	=	14.7 psi
1 bar	=	10 ⁵ Pa
	=	100 kPa

Energy		
1 cal	=	4.184 J
1 L atm	=	101.325 J
1 Cal	=	1 kcal
1 hp	=	746 W
1 eV	=	1.602×10^{-19} J

Various Physical Constants

property	symbol	value
universal gas constant	R	8.314 J mol ⁻¹ K ⁻¹ 62.36 L torr mol ⁻¹ K ⁻¹ 0.08206 L atm mol ⁻¹ K ⁻¹ 1.987 cal mol ⁻¹ K ⁻¹
Planck's constant	h	6.626×10^{-34} J s 4.136×10^{-15} eV s
Planck's reduced constant	$h/2\pi$	1.054×10^{-34} J s 6.582×10^{-16} eV s
Boltzmann constant	k_B	1.38×10^{-23} J K ⁻¹
Stefan-Boltzmann	σ	5.67×10^{-8} W m ⁻² K ⁻⁴
speed of light	c	3.00×10^8 m s ⁻¹
speed of sound (at 20°C)	v_{air}	343 m s ⁻¹
acceleration of gravity	g	9.80 m s ⁻²
gravitational constant	G	6.67×10^{-11} N m ² kg ⁻²
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
elementary charge	e	1.602×10^{-19} C
Faraday	F	96485 C mol ⁻¹
Coulomb's law constant	k	8.988×10^9 N m ² C ⁻²
Rydberg constant	R	2.178×10^{-18} J

Some Other Conversion Factors

1 in	=	2.54 cm
1 lb	=	453.6 g
1 mi	=	5280 ft = 1.609 km
1 gal	=	4 quarts = 231 in ³ = 3.785 L

property	symbol	value
electron rest mass	m_e	9.11×10^{-31} kg 0.000549 u 0.511 MeV c ⁻²
proton mass	m_p	1.6726×10^{-27} kg 1.00728 u 938.3 MeV c ⁻²
neutron mass	m_n	1.6749×10^{-27} kg 1.008665 u 939.6 MeV c ⁻²
atomic mass unit	u	1.6605×10^{-27} kg 931.5 MeV c ⁻²
earth mass		5.972×10^{24} kg
earth radius		6.371×10^6 m
moon mass		7.348×10^{22} kg
sun mass		1.989×10^{30} kg
distance earth-moon		3.844×10^8 m
distance earth-sun		1.496×10^{11} m
permittivity of free space	ϵ_0	8.85×10^{-12} F m ⁻¹
permeability of free space	μ_0	$4\pi \times 10^{-7}$ T m A ⁻¹

Some Average Bond Energies (kJ/mol)

C-H	413	C-C	346	C-Cl	339	C-N	305
O-H	463	C=C	602	C-Br	285	N=N	418
N-H	391	C≡C	835	O=O	498	H-H	436
C-O	358	C=O	799	C≡O	1072	Br-Br	193
H-Cl	432	S-H	347	N≡N	945	Cl-Cl	242
H-Br	366	H-I	299	C≡N	887	I-I	151

Biology Questions (1 – 20)

1. Monotremes and marsupials are two main clades of the _____ lineage.
 - a. eutherian
 - b. animal
 - c. primate
 - d. aves
 - e. mammal
2. The central nervous system in vertebrates consists of the _____ and _____.
 - a. sensory neurons and motor neurons
 - b. nerves and ganglia
 - c. cerebrum and cerebellum
 - d. brain and spinal cord
 - e. white matter and gray matter
3. “What is the concentration of protons in a solution with a pH of 2?
 - a. 10^{-2}
 - b. 10^{-6}
 - c. 10^{-12}
 - d. 10^{-14}
 - e. 10^{-20}
4. The difference in the number of phosphate groups between ADP and ATP is
 - a. 0
 - b. 1
 - c. 2
 - d. 3
 - e. 4
5. How many saturated fatty acids could be found in a fat molecule?
 - a. 0
 - b. 2
 - c. 3
 - d. 4
 - e. Cannot be determined
6. Populations separated by a geographical barrier are _____.
 - a. divergent
 - b. sympatric
 - c. geographic
 - d. allopatric
 - e. allologous
7. The process by which one cell divides into two identical cells is called
 - a. binary fission
 - b. mitosis
 - c. meiosis
 - d. A and B
 - e. A, B, and C
8. The integumentary system is made up of
 - a. heart, blood and blood vessels
 - b. smooth, skeletal, cardiac muscles
 - c. nose, throat, and lung
 - d. skin, hair and nails
 - e. mouth, stomach, and digestive tract
9. Which of the following are the main human blood types?
 - a. A, B and O
 - b. AA, BB and OO
 - c. A, B, AB and O
 - d. A, B, AO and O
 - e. A, B, O and i
10. The physical characteristic of an organism is the
 - a. fitness
 - b. phenotype
 - c. genotype
 - d. allele
 - e. type

11. The tissue that brings water and nutrients from the soil to the leaves of a plant is called
 - a. stem tissue
 - b. root tissue
 - c. phloem
 - d. fibers
 - e. xylem
12. Autotrophs use which of the following to power cellular work?
 - a. sunlight
 - b. CO₂
 - c. ATP
 - d. H₂O
 - e. cellulose
13. Which of the following are methods of pollination in plants?
 - a. wind
 - b. water
 - c. animals
 - d. A and C
 - e. A, B and C
14. The primary photosynthetic pigment is
 - a. chloroplast
 - b. chlorophyll
 - c. sunlight energy
 - d. thylakoid
 - e. a photon
15. Which of the following is NOT a basic component of cell theory?
 - a. All cells arise from other cells
 - b. Cells are the most basic unit of life.
 - c. Viruses are the smallest type of cell.
 - d. All living organisms are composed of cells.
16. A dihybrid cross follows the inheritance of _____ traits.
 - a. observable
 - b. heritable
 - c. multiple
 - d. genetic
 - e. all of the above
17. All groups of animals have a nervous system except:
 - a. insects
 - b. worms
 - c. sponges
 - d. mollusks
 - e. arthropods
18. The cuticle is a waxy layer found on leaves and shoots of terrestrial plants. What is the function of the cuticle?
 - a. increase water absorption
 - b. decrease water evaporation
 - c. increase photosynthesis
 - d. increase leaf temperature
 - e. decrease herbivory
19. Bacteria that have an outer membrane over the cell wall are known as
 - a. anaerobes
 - b. gram-positive bacteria
 - c. gram-negative bacteria
 - d. Archaea
 - e. capsular bacteria
20. Which of the following contain all the others?
 - a. Genus
 - b. Phylum
 - c. Class
 - d. Family
 - e. Domain

Chemistry Questions (21 – 40)

21. What is the name for the compound $\text{Sr}(\text{ClO})_2$?
- strontium chlorite
 - serium dichlorate
 - strontium hypochlorite
 - strontium perchlorate
 - soresium chlorite
22. What is the chemical formula for potassium thiosulfate?
- KThSO_3
 - PS_3O_6
 - PoSO_4
 - K_2SO_3
 - $\text{K}_2\text{S}_2\text{O}_3$
23. What is the molar mass of $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$?
- 129.8 g/mol
 - 237.9 g/mol
 - 14000 g/mol
 - 202.5 g/mol
 - 147.8 g/mol
24. Epsom salt is the hydrated form of MgSO_4 . What is the percentage of magnesium by mass in MgSO_4 ?
- 13.9 %
 - 16.7 %
 - 20.2 %
 - 23.3 %
 - 25.3 %
25. Which of the following ions will have the smallest radius?
- Mg^{2+}
 - F^-
 - Na^+
 - O^{2-}
 - K^+
26. Balance the following reaction with whole numbers as coefficients.
- $$\text{C}_3\text{H}_{10}\text{N}_2(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{NO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$$
- What are the respective coefficients?
- $3 : 21 \rightarrow 9 : 6 : 15$
 - $1 : 8 \rightarrow 3 : 2 : 5$
 - $1 : 7 \rightarrow 3 : 4 : 8$
 - $2 : 15 \rightarrow 6 : 4 : 10$
 - $2 : 15 \rightarrow 3 : 2 : 5$
27. What is the oxidation number for nitrogen in the toxic gas NO_2 ?
- 3
 - +4
 - +5
 - 2
 - +3
28. Balance the following equation.
- $$\text{Al}_2\text{O}_3 + \text{HF} \rightarrow \text{AlF}_3 + \text{H}_2\text{O}$$
- What is the maximum number of grams of AlF_3 that can be formed when 50 g of Al_2O_3 and 55 g of HF completely react?
- 93.1 g
 - 82.4 g
 - 41.2 g
 - 77.0 g
 - 69.7 g
29. KNO_3 is a major component of fertilizer and can be made by the following reaction
- $$4 \text{KCl} + 4 \text{HNO}_3 + \text{O}_2 \rightarrow 4 \text{KNO}_3 + 2 \text{Cl}_2 + 2 \text{H}_2\text{O}$$
- How many pounds of HNO_3 are needed to make 850 lbs of KNO_3 ?
- 530 lbs
 - 627 lbs
 - 1284 lbs
 - 479 lbs
 - 721 lbs
30. What is the electron configuration of zirconium (Zr) ?
- $[\text{Kr}] 5s^2 5d^2$
 - $[\text{Kr}] 5s^0 4d^4$
 - $[\text{Kr}] 5s^2 4d^2$
 - $[\text{Kr}] 5s^2 4d^{10} 5p^2$

31. A single photon delivers 2.209×10^{-19} joules of energy to a molecule. What is the wavelength of this photon?
- A) 325 nm
B) 450 nm
C) 600 nm
D) 725 nm
E) 900 nm
32. What is the $\angle\text{COC}$ bond angle in dimethyl ether, CH_3OCH_3 ?
- A) 120°
B) 109.5°
C) 180°
D) 90°
E) 60°
33. Which of the following molecules is polar?
- A) SF_4
B) CF_4
C) SF_6
D) PF_5
E) BF_3
34. An endothermic reaction is at equilibrium. As the reaction is slowly cooled, which of the following must be true about the reactants and products?
- A) the products will increase in concentration
B) the reactants will increase in concentration
C) the concentrations of both reactants and products will remain constant
35. 25.0 mL of sulfuric acid (H_2SO_4) is titrated to the complete end point with 31.05 mL of 0.00765 M KOH. What is the concentration of the sulfuric acid?
- A) 0.00950 M
B) 0.00725 M
C) 0.00350 M
D) 0.00475 M
E) 0.00525 M
36. A 10 liter gas container is filled to a pressure of 500 torr with CO_2 gas at 25°C . How many grams of CO_2 are in the container?
- A) 0.269 g
B) 7.36 g
C) 11.8 g
D) 16.2 g
E) 38.4 g
37. Gas A has a molar mass of 40 g/mol and Gas B has a molar mass of 60 g/mol. If equal masses of the two gases are put into a container what will be the mole fraction of Gas B in the container?
- A) $1/5$
B) $2/5$
C) $3/5$
D) $1/3$
E) $2/3$
38. The molar solubility of $\text{Ni}(\text{OH})_2$ is 3.42×10^{-6} mol/L. What is the value of K_{sp} for this compound ?
- A) 1.2×10^{-11}
B) 4.0×10^{-17}
C) 3.4×10^{-6}
D) 1.4×10^{-22}
E) 1.6×10^{-16}
39. What is the pH of a 1.5×10^{-5} M solution of $\text{Sr}(\text{OH})_2$?
- A) 4.52 C) 10.25 E) 9.48
B) 4.82 D) 9.18
40. Which of the equations shown is equal to the equilibrium constant, K_p , for the following gas phase chemical reaction?
- $$\text{A(g)} + 2 \text{B(g)} \rightleftharpoons 3 \text{C(g)}$$
- A) $\frac{P_{\text{C}}^3}{P_{\text{A}} P_{\text{B}}^2}$ C) $\frac{P_{\text{C}}}{P_{\text{A}} P_{\text{B}}}$
B) $\frac{P_{\text{A}} P_{\text{B}}^2}{P_{\text{C}}^3}$ D) $\frac{3P_{\text{C}}}{P_{\text{A}} 2P_{\text{B}}}$

Physics Questions (41 – 60)

On this contest $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors in the $+x$, $+y$ and $+z$ directions respectively, and unless otherwise indicated, you may neglect air resistance in every problem.

41. According to Tyson, which astronomer claimed that Mars was inhabited by an intelligent civilization?
A) Edmond Halley
B) Sir William Herschel
C) Percival Lowell
D) Clyde W. Tombaugh
E) H. G. Wells
42. According to Tyson, initially Pluto was assumed to have a size and mass similar to which planet?
A) Jupiter
B) Mars
C) Neptune
D) Saturn
E) Uranus
43. According to Tyson, what event(s) precipitated a recalculation of planetary masses of Jupiter, Saturn, Uranus, and Neptune? This event(s) fixed the orbital parameters of the planets and thus “killed” the need for a Planet X.
A) the development of better telescopes
B) the discovery of the asteroid belt
C) the discovery of Pluto’s moon Charon
D) the recalculation of Pluto’s mass using its brightness
E) the Voyager flybys
44. According to Tyson, what planet was reclassified as the largest asteroid?
A) Ceres
B) Delta Geminorum
C) Juno
D) Pallas
E) Vesta
45. A motorcycle traveling on a straight road speeds up at a constant acceleration of 5.0 m/s^2 . If the initial speed of the motorcycle is 25 m/s , then what is the speed of the motorcycle after 7.0 s ?
A) 10 m/s
B) 25 m/s
C) 32 m/s
D) 35 m/s
E) 60 m/s
46. A boy holds his BB gun 40.0 inches above the ground to shoot. The boy shoots horizontally at a target but misses the target and his BB hits the ground. If the initial speed of the BB is 1000 ft/s , then what is the horizontal range of the BB?
A) 63.2 m
B) 139 m
C) 207 m
D) 305 m
E) 670 m
47. While redecorating you have to push a heavy trunk across a room at a constant velocity. In an epiphany you decide to flip the trunk onto its smaller side so that you reduce the amount of surface area of the trunk in contact with the floor by a factor of one quarter. If you continue to move the trunk with the same constant velocity as before, then the horizontal force you apply during the motion is
A) a quarter of the previous horizontal force
B) half of the previous horizontal force
C) the same as the previous horizontal force
D) twice the previous horizontal force
E) four times the previous horizontal force
48. A car travels around a flat (unbanked) curve that has a radius of 325 m . If the coefficient of static friction between the tires and the road is only 0.650 (due to wet conditions), then what is the maximum speed of the car if it is to travel along the curve without slipping or skidding?
A) 45.5 m/s
B) 69.3 m/s
C) 211 m/s
D) 1460 m/s
E) 6530 m/s

49. A hockey puck on a frictionless ice rink has a horizontal force applied to it that can be described by the following potential-energy function: $U(x,y) = (3/2)k(x^2 + 2y^2)$, where x and y are horizontal coordinates. What is the x -component of the force?
- $-3kx$
 - $-(3/4)kx$
 - $-(1/2)kx^3$
 - $+(3/4)kx$
 - $+3kx$
50. Which of the following is NOT true of the amount of work done by a conservative force?
- The amount of work done on a path that is reversed is the negative of the amount of work done on the path in the original direction.
 - The amount of work done depends on the path taken from the starting to the ending points of the path.
 - The amount of work done only depends on the starting and ending points of the path taken.
 - When the starting and ending points are the same then the amount of work done is zero.
 - The amount of work done can be expressed as the difference between the initial and final values of a scalar potential energy function.
51. A solid sphere, a solid cylinder, a hollow sphere, and a hollow cylinder have the same mass and the same diameters. If all objects are released from rest at the same time and roll without slipping down an incline, then which object will reach the bottom of the incline first?
- all objects will reach the bottom of the incline at the same time
 - solid sphere
 - solid cylinder
 - hollow sphere
 - hollow cylinder
52. While fixing a tire on your car you need to use a jack. To be able to move the jack handle you need to put a “cheater” on the handle to extend the length of the handle and then apply your full weight of 750 N. The handle plus cheater are 0.750 m long while the cheater and handle make a 15.0° angle with the horizontal. What is the magnitude of the torque you applied to the jack?
- 146 N·m
 - 543 N·m
 - 562 N·m
 - 1040 N·m
 - 3860 N·m
53. A 20.0 kg solid sculpture made of pure gold (with density $19.3 \times 10^3 \text{ kg/m}^3$) was located at the bottom of a lake and is recovered by lifting it to the surface by a cable. What is the magnitude of the force of tension applied to the sculpture by the cable while it is being lifted at a constant speed from the lake bottom with the sculpture is still completely submerged?
- 10.3 N
 - 20.6 N
 - 186 N
 - 196 N
 - 206 N
54. Water enters your house through a pipe with an inner diameter of 3.0 cm with a gauge pressure of $5.0 \times 10^5 \text{ Pa}$. The water continues upstairs to the tub through a pipe of inner diameter of 2.0 cm. If the bath is 2.9 m above the ground and the inlet pipe has a flow speed of 3.0 m/s then how long does it take to fill the tub with 150 L of water?
- $3.2 \times 10^{-1} \text{ s}$
 - $7.1 \times 10^1 \text{ s}$
 - $3.2 \times 10^2 \text{ s}$
 - $1.0 \times 10^3 \text{ s}$
 - $6.4 \times 10^4 \text{ s}$

55. What is the internal energy of 4.00 mol of krypton that is kept at 20.0°C?
- 4.14×10^{-22} J
 - 2.02×10^{-21} J
 - 2.95×10^2 J
 - 1.00×10^3 J
 - 1.46×10^4 J
56. Two point charges $q_1 = -25.0 \mu\text{C}$ and $q_2 = +25.0 \mu\text{C}$ are placed 20.0 cm apart. What is the magnitude of the net electric field at the point halfway between the two charges?
- 0.00 N/C
 - 5.62×10^6 N/C
 - 1.12×10^7 N/C
 - 2.25×10^7 N/C
 - 4.50×10^7 N/C
57. A nickel wire has a diameter of 2.50 mm and a cross sectional area of $9.50 \times 10^{-6} \text{ m}^2$. The wire has a resistance of 0.750 Ω at 20.0°C. If the temperature coefficient of resistance for nickel is 0.005866 K^{-1} , then what is the resistance of the wire at 80.0°C? You may assume that the temperature coefficient remains constant over the temperature range in this problem.
- 0.490 Ω
 - 0.950 Ω
 - 1.01 Ω
 - 1.10 Ω
 - 1.19 Ω
58. Which of the following is NOT true of the magnetic field of the Earth?
- The Earth's geomagnetic south pole (the magnetic pole in the southern hemisphere) is currently a magnetic north pole (as found on a bar magnet).
 - The location of Earth's geomagnetic poles move by as much as 40 km/year.
 - The Earth's geomagnetic poles are at different locations than the geographic poles (the locations where the Earth's axis of rotation meets the surface of the Earth).
 - The Earth's magnetic field is generated by electric currents in the conductive material of its mantle.
 - The strength of the Earth's magnetic field has been decreasing at a rate of about 6.3% per century over the last two centuries.
59. What is the self-inductance of a toroidal solenoid given that the solenoid has a cross-sectional area of 12.0 cm^2 , a radius of 25 cm and is comprised of 300 turns of wire?
- 5.76×10^{-7} H
 - 8.64×10^{-5} H
 - 5.43×10^{-4} H
 - 1.73×10^{-4} H
 - 2.38×10^{-2} H
60. A converging lens has a focal length of +20 cm. What is the lateral magnification of an object that is located 40 cm in front of the lens? You may approximate the lens as a thin lens.
- 1
 - 2/3
 - 1/3
 - 1
 - 2

UIL HIGH SCHOOL SCIENCE CONTEST

Contestant Answer Sheet

Contestant # _____

9 10 11 12
CONTESTANT GRADE LEVEL

A 2A 3A 4A 5A
CONFERENCE

PLEASE WRITE ANSWERS WITH CAPITAL LETTERS

- | | | |
|-----------|-----------|-----------|
| 1. _____ | 21. _____ | 41. _____ |
| 2. _____ | 22. _____ | 42. _____ |
| 3. _____ | 23. _____ | 43. _____ |
| 4. _____ | 24. _____ | 44. _____ |
| 5. _____ | 25. _____ | 45. _____ |
| 6. _____ | 26. _____ | 46. _____ |
| 7. _____ | 27. _____ | 47. _____ |
| 8. _____ | 28. _____ | 48. _____ |
| 9. _____ | 29. _____ | 49. _____ |
| 10. _____ | 30. _____ | 50. _____ |
| 11. _____ | 31. _____ | 51. _____ |
| 12. _____ | 32. _____ | 52. _____ |
| 13. _____ | 33. _____ | 53. _____ |
| 14. _____ | 34. _____ | 54. _____ |
| 15. _____ | 35. _____ | 55. _____ |
| 16. _____ | 36. _____ | 56. _____ |
| 17. _____ | 37. _____ | 57. _____ |
| 18. _____ | 38. _____ | 58. _____ |
| 19. _____ | 39. _____ | 59. _____ |
| 20. _____ | 40. _____ | 60. _____ |

BIOLOGY SCORE

CHEMISTRY SCORE

PHYSICS SCORE

OVERALL SCORE:

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY**

INVITATIONAL B • 2015

- | | | |
|-------|-------|-------|
| 1. E | 21. C | 41. C |
| 2. D | 22. E | 42. C |
| 3. A | 23. B | 43. E |
| 4. B | 24. C | 44. A |
| 5. C | 25. A | 45. E |
| 6. D | 26. D | 46. B |
| 7. D | 27. B | 47. C |
| 8. D | 28. D | 48. A |
| 9. C | 29. A | 49. A |
| 10. B | 30. C | 50. B |
| 11. E | 31. E | 51. B |
| 12. C | 32. B | 52. B |
| 13. E | 33. A | 53. C |
| 14. B | 34. B | 54. B |
| 15. C | 35. D | 55. E |
| 16. E | 36. C | 56. E |
| 17. C | 37. B | 57. C |
| 18. B | 38. E | 58. D |
| 19. C | 39. E | 59. B |
| 20. E | 40. A | 60. A |

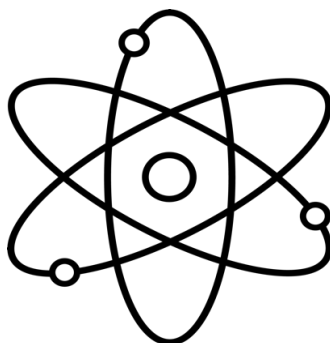
PHYSICS KEY for Science Contest • Invitational B • 2015

41. (C) "In his early years, Percival Lowell indulged a fanatical, even delusional fascination with Mars, claiming that intelligent civilizations were in residence there, digging networks of canals to channel water from the polar ice caps to the cities."
42. (C) "Pluto was first presumed to be of commensurate rank in size and mass with Neptune, itself about 18 times Earth's mass."
43. (E) "Standish used the updated mass estimates for Jupiter, Saturn, Uranus, and Neptune that had become available from the Voyager flybys; ..."
44. (A) "Ceres was discovered first because it's the brightest and largest of its class. At twice the mass of all other asteroids combined, of which there are hundreds of thousands known, Ceres swiftly went from the smallest in the class of planets to the largest in the class of asteroid."
45. (E) By the kinematic equation: $v = v_0 + at = (+25) + (+5)(7) = 60 \text{ m/s}$
46. (B) By the kinematic equations: $y = y_0 + v_{0y}t + \frac{1}{2}a_yt^2 \Rightarrow t = [2(40 \cdot 2.54/100)/9.8]^{1/2} = 0.4554 \text{ s}$,
thus by $x = x_0 + v_{0x}t = 0 + (1000 \cdot 12 \cdot 2.54/100)(0.4554) = 139 \text{ m}$
47. (C) To maintain a constant velocity during the motion your applied force must be equal in magnitude and opposite in direction to the force of kinetic friction. The force of kinetic friction is independent of the area of contact between the two surfaces, and thus so is your applied force.
48. (A) From the FBD & N2L: (vertically) $+F_N - F_G = 0 \Rightarrow F_N = mg$ (radially) $+F_{\text{frs}} = ma_c = mv^2/r$ & with $F_{\text{frs}} \leq \mu_s F_N = \mu_s mg \Rightarrow \mu_s mg \leq mv^2/r \Rightarrow v \leq [\mu_s gr]^{1/2} = [(0.65)(9.8)(325)]^{1/2} = 45.5 \text{ m/s}$
49. (A) $F = -\nabla U \Rightarrow F_x = -\partial U/\partial x = -3kx$
50. (B) The amount of work done by a conservative force is *independent* of the path taken from the starting to the ending points of the path.
51. (B) The object with the highest translation speed of the center of mass (and thus the first to reach the bottom of the incline) will be the one with the lowest moment of inertia which is solid sphere.
52. (B) By $\tau = rF \sin\theta = (0.75)(750) \sin(15^\circ + 90^\circ) = 543 \text{ N}\cdot\text{m}$
53. (C) From the FBD & N2L: (vertically) $+F_T + F_B - F_G = 0 \Rightarrow F_T = F_G - F_B = mg - \rho_{\text{fluid}} V_{\text{object submerged}} g$
and with $V = m/\rho \Rightarrow F_T = (20)[1 - (1000/19300)](9.8) = 186 \text{ N}$
54. (B) For an incompressible fluid the volume flow rate, Q , remains constant at any point along a pipe, thus $Q = vA \cos\theta = \Delta V/\Delta t \Rightarrow \Delta t = [\Delta V]/[vA \cos\theta] = [150/1000]/[3(\pi \cdot 0.015^2) \cos 0^\circ] = 7.1\text{E}+1 \text{ s}$
55. (E) For a monatomic ideal gas: $E_{\text{int}} = (3/2)nRT = (3/2)(4)(8.314)(20+273.15) = 1.46\text{E}+4 \text{ J}$
56. (E) $E_{\text{net}} = E_1 + E_2 = 2E = 2k|q|/r^2 = 2(8.99\text{E}+9)(25\text{E}-6)/(0.1)^2 = 4.50\text{E}+7 \text{ N/C}$
57. (C) By $R = R_0 [1 + \alpha\Delta T] = (0.75)[1 + (0.005866)(20)] = 1.01 \Omega$
58. (D) The Earth's magnetic field is generated by electric currents in the conductive material of its core (not its mantle). The mantle is a predominately solid, silicate rocky shell that encloses a hot core rich in iron and nickel where convective currents act as the source of the Earth's magnetic field.
59. (B) By definition: $L = N\Phi_B/I = N\{BA \cos 0^\circ\}/I = N\{[(\mu_0 NI)/(2\pi r)]A\}/I = [\mu_0 N^2 A]/[2\pi r] = [(4\pi\text{E}-7)(300)^2(0.0012)]/[2\pi(0.25)] = 8.64\text{E}-5 \text{ H}$
60. (A) By $1/f = 1/d_o + 1/d_i = 1/20 = 1/40 + 1/d_i \Rightarrow d_i = +40 \text{ cm}$. Thus $m = -d_i/d_o = -(40)/(40) = -1$



SCIENCE

District 1 • 2015



GENERAL DIRECTIONS:

- DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
- Ninety minutes should be ample time to complete this contest, but since it is not a race, contestants may take up to two hours. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet.
- You may place as many notations as you desire anywhere on the test paper except on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- On the back of this page is printed a copy of the periodic table of the elements. You may wish to refer to this table in answering the questions, and if needed, you may use the atomic weights and atomic numbers from the table. Other scientific relationships are listed also.
- Silent hand-held calculators that do not need external wall plugs may be used. Graphing calculators that do not have built-in or stored functionality that provides additional scientific information are allowed. Small hand-held computers are not permitted. Calculators that accept memory cards or memory sticks are not permitted. Each contestant may bring one spare calculator. All memory must be cleared.

SCORING:

All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for an incorrect answer.

Periodic Table of the Elements

1A 1		2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.008																		2 He 4.003
3 Li 6.941	4 Be 9.012												5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12		13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41		31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41		49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59		81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (285)							

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Some Standard Properties of Water

property	symbol	value
density of water	ρ_{water}	1.000 g cm ⁻³
density of ice	ρ_{ice}	0.9167 g cm ⁻³
specific heats		
ice	C_{ice}	2.09 J g ⁻¹ K ⁻¹
water	C_{water}	4.184 J g ⁻¹ K ⁻¹
steam	C_{steam}	2.03 J g ⁻¹ K ⁻¹
heat of fusion	ΔH_{fus} or L_f	334 J g ⁻¹
heat of vaporization	ΔH_{vap} or L_v	2260 J g ⁻¹
index of refraction	n	1.33
autoionization	K_w	1.0×10^{-14}

Pressure		
1 atm	=	760 torr
	=	101325 Pa
	=	14.7 psi
1 bar	=	10 ⁵ Pa
	=	100 kPa

Energy		
1 cal	=	4.184 J
1 L atm	=	101.325 J
1 Cal	=	1 kcal
1 hp	=	746 W
1 eV	=	1.602×10^{-19} J

Various Physical Constants

property	symbol	value
universal gas constant	R	8.314 J mol ⁻¹ K ⁻¹ 62.36 L torr mol ⁻¹ K ⁻¹ 0.08206 L atm mol ⁻¹ K ⁻¹ 1.987 cal mol ⁻¹ K ⁻¹
Planck's constant	h	6.626×10^{-34} J s 4.136×10^{-15} eV s
Planck's reduced constant	$h/2\pi$	1.054×10^{-34} J s 6.582×10^{-16} eV s
Boltzmann constant	k_B	1.38×10^{-23} J K ⁻¹
Stefan-Boltzmann	σ	5.67×10^{-8} W m ⁻² K ⁻⁴
speed of light	c	3.00×10^8 m s ⁻¹
speed of sound (at 20°C)	v_{air}	343 m s ⁻¹
acceleration of gravity	g	9.80 m s ⁻²
gravitational constant	G	6.67×10^{-11} N m ² kg ⁻²
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
elementary charge	e	1.602×10^{-19} C
Faraday	F	96485 C mol ⁻¹
Coulomb's law constant	k	8.988×10^9 N m ² C ⁻²
Rydberg constant	R	2.178×10^{-18} J

Some Other Conversion Factors

1 in	=	2.54 cm
1 lb	=	453.6 g
1 mi	=	5280 ft = 1.609 km
1 gal	=	4 quarts = 231 in ³ = 3.785 L

property	symbol	value
electron rest mass	m_e	9.11×10^{-31} kg 0.000549 u 0.511 MeV c ⁻²
proton mass	m_p	1.6726×10^{-27} kg 1.00728 u 938.3 MeV c ⁻²
neutron mass	m_n	1.6749×10^{-27} kg 1.008665 u 939.6 MeV c ⁻²
atomic mass unit	u	1.6605×10^{-27} kg 931.5 MeV c ⁻²
earth mass		5.972×10^{24} kg
earth radius		6.371×10^6 m
moon mass		7.348×10^{22} kg
sun mass		1.989×10^{30} kg
distance earth-moon		3.844×10^8 m
distance earth-sun		1.496×10^{11} m
permittivity of free space	ϵ_0	8.85×10^{-12} F m ⁻¹
permeability of free space	μ_0	$4\pi \times 10^{-7}$ T m A ⁻¹

Some Average Bond Energies (kJ/mol)

C-H	413	C-C	346	C-Cl	339	C-N	305
O-H	463	C=C	602	C-Br	285	N=N	418
N-H	391	C≡C	835	O=O	498	H-H	436
C-O	358	C=O	799	C≡O	1072	Br-Br	193
H-Cl	432	S-H	347	N≡N	945	Cl-Cl	242
H-Br	366	H-I	299	C≡N	887	I-I	151

Biology Questions (1 – 20)

1. Catabolic pathways release energy by the _____ of organic molecules.
 A) synthesis
 B) reduction
 C) anabolism
 D) oxidation
 E) generation
2. DNA _____ is a process used in biotechnology that results in multiple copies of a gene.
 A) cloning
 B) replication
 C) transcription
 D) chain reactions
 E) multiplication
3. Animals and fungi are similar in that they are both _____.
 A) multicellular
 B) made of chitin
 C) diploid for the majority of the life cycle
 D) heterororphs
 E) autotrophs
4. Which of the following is NOT a function of a protein found in the plasma membrane?
 A) Cellular joining
 B) Catalyzing chemical reactions
 C) Transport of ions
 D) Peptide synthesis
 E) Protein secretion
5. The molecular connectors between DNA and polypeptides is/are _____.
 A) histone proteins
 B) RNA's
 C) the nucleus
 D) the rough endoplasmic reticulum
 E) peptide bonds
6. How many ester linkages between carboxylic acids and glycerol would be found in a phospholipid?
 A) 0
 B) 1
 C) 2
 D) 3
 E) cannot be determined from the information provided.
7. The chromosomal complement found in human male gamete contains
 A) 23 autosomes and an X chromosome
 B) 23 autosomes and a Y chromosome
 C) 23 autosomes and either an X or a Y chromosome
 D) 22 autosomes and both an X and Y chromosome
 E) none of the above
8. Bony fishes (Osteichthyes) are divided into two main clades: _____ and _____.
 A) salt water, fresh water
 B) ray finned, lobe finned
 C) oviparous, viviparous
 D) chordates, craniates
 E) scaled, smooth
9. How many different kinds of amino acids could be found in a protein?
 A) 4 different kinds: A, T, G, C
 B) 5 different kinds: A, T, G, C, U
 C) 3 different kinds: phosphate, sugar and base
 D) 20 different kinds:
 GAVLIMFWPSTCYNQDEKRRH
 E) 61 different kinds: 64 codons less the stop codons
10. The process by which animals control solute concentrations and balance water gain and loss is called _____.
 A) excretion
 B) osmosis
 C) osmolarity
 D) osmoregulation
 E) excregulation

11. Potential energy in biological systems can be found in
 - A) Electrical bonds
 - B) Covalent bonds
 - C) Thermal bonds
 - D) A & B
 - E) A, B and C
12. Crossing over occurs during which phase of meiosis?
 - A) prophase
 - B) prophase I
 - C) prophase II
 - D) metaphase I
 - E) metaphase
13. According to a phylogenetic tree, which of the following is NOT a reptile?
 - A) eagle
 - B) snake
 - C) turtle
 - D) frog
 - E) crocodile
14. Terrestrial vertebrates all share which of the following in common?
 - A) They are all tetrapods.
 - B) They are all amniotic.
 - C) They are all adapted to live on land.
 - D) They are all protostomes.
 - E) They all give birth to live young.
15. Homo sapiens migrated out of Africa about 100,000 years ago. Which one of the following species might the early H. sapiens have encountered?
 - A) Australopithecus afarensis
 - B) Homo habilis
 - C) Homo neanderthalensis
 - D) Australopithecus africanus
 - E) Tyrannosaurus rex
16. The complete metamorphosis of arthropods includes which of the following stages?
 - A) zygote, larva, cocoon, adult
 - B) Pupa, larva, cocoon, adult
 - C) Larva, pupa, adult
 - D) Larva, chrysalis, cocoon, adult
 - E) Pupa, cocoon, adult
17. Plants adapted to live on land have a waxy coating on their leaves called the cuticle. Which is the main function of the cuticle?
 - A) to make the plant distasteful to herbivores
 - B) to allow germination to occur more quickly
 - C) to attract pollinators
 - D) to avoid dessication
 - E) to allow leaves to float on water
18. Which of the following were primarily responsible for the accumulation of free oxygen on the early Earth?
 - A) archaea
 - B) photosynthetic bacteria
 - C) photosynthetic eukaryote
 - D) chemosynthetic bacteria
 - E) chloroplasts
19. The two main divisions of the skeletal system are
 - A) bone and bone marrow
 - B) cranial and subcranial
 - C) axial and appendicular
 - D) cranial and vertebral
 - E) axial and vertebral
20. The rate at which a substance diffuses is affected by the
 - A) temperature of the solution.
 - B) concentration gradient in the system.
 - C) presence of ATP.
 - D) A and B
 - E) A, B and C

Chemistry Questions (21 – 40)

21. What is the name of the compound, $\text{Sn}(\text{C}_2\text{H}_3\text{O}_2)_2$?
- antimony(II) oxalate
 - tin carbonate
 - tin(II) diacetic acid
 - antimony acetate
 - tin(II) acetate
22. Balance the following reaction with whole numbers as coefficients.
- $$\text{PCl}_3(\text{s}) + \text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_3 + \text{HCl}$$
- What are the respective coefficients?
- 2 : 6 : 2 : 7
 - 1 : 2 : 2 : 5
 - 1 : 3 : 1 : 3
 - 3 : 6 : 3 : 9
 - 1 : 1 : 1 : 3
23. A laboratory bottle is labeled 15% KOH with a density of 1.122 g/mL. What is the molarity of the KOH?
- 3.00 M
 - 2.67 M
 - 2.55 M
 - 3.12 M
 - less than 2.20 M
24. When a metal reacts with a non-metal, the resulting compound is usually _____.
- a covalent compound
 - a polymer
 - a bimetallic compound
 - an ionic compound
 - a protein
25. What is the formal charge on the nitrogen in the ion, NH_4^+ ?
- +2
 - 2
 - 1
 - +1
 - 0
26. A sample of hydrogen gas has a volume of 725 mL at STP. What would be the volume of this sample of gas at 143°C and 280 torr?
- 3.00 L
 - 2.00 L
 - 3.50 L
 - 2.50 L
 - 4.00 L
27. A salt is known to have a negative value for $\Delta H_{\text{solution}}$. A sample of the salt is dissolved into some water but some of the salt remains undissolved. Which of the following procedures will help to dissolve the rest of the salt?
- heat the solution
 - cool the solution
 - add more water
 - add more salt
- I and III
 - I and IV
 - II and III
 - II and IV
 - only IV
28. A saturated solution of barium phosphate ($K_{\text{sp}} = 5.8 \times 10^{-39}$) has a barium concentration of 2.5×10^{-7} M. What is the concentration of phosphate in this solution?
- 1.5×10^{-16} M
 - 6.1×10^{-10} M
 - 3.0×10^{-13} M
 - 8.8×10^{-9} M
 - 5.4×10^{-14} M
29. Which solution listed has the highest boiling point?
- 2.5 *m* NaCl
 - 1.5 *m* CaCl_2
 - 1.0 *m* Na_3PO_4
 - 2.0 *m* KBr
 - 1.5 *m* $\text{Al}(\text{NO}_3)_3$
30. Consider a system at 300 K with the following reaction occurring:
- $$2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{H}_2\text{O}(\text{l})$$
- What is the amount of work done? Is the work done *on* the system or *by* the system?
- 7.5 kJ of work done on the system
 - 2.5 kJ of work done by the system
 - no work done on or by the system
 - 7.5 kJ of work done by the system
 - 2.5 kJ of work done on the system

31. Methanol (CH_3OH) has the following phase data:

$$\Delta H_{\text{fusion}} = 2.20 \text{ kJ/mol} \quad T_{\text{fp}} = -97^\circ\text{C}$$

$$\Delta H_{\text{vaporization}} = 37.6 \text{ kJ/mol} \quad T_{\text{bp}} = 65^\circ\text{C}$$

$$C_{\text{p,liquid}} = 79.5 \text{ J/mol K}$$

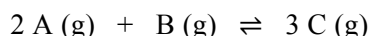
How much heat must be removed from 96.1 g of methanol at 25°C in order to change it to frozen (solid) methanol at -97°C ?

- A) 29.1 kJ C) 9.70 kJ E) 11.9 kJ
B) 35.7 kJ D) 47.3 kJ

32. Which example has $\Delta S < 0$ for the change given?

- A) sugar cube dissolves into water
B) ethanol liquid evaporates
C) liquid H_2O freezes into ice
D) liquid H_2O is electrolyzed into H_2 and O_2
E) benzene liquid is heated from 25°C to 30°C

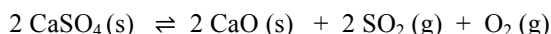
33. Consider the gas phase equilibrium:



If the partial pressure of C is doubled, what happens to the reaction quotient, Q ?

- A) it drops to $1/8^{\text{th}}$ its original value
B) it increases by a factor of 2
C) Q will not change
D) it increases by a factor of 8
E) it drops to $1/4^{\text{th}}$ its original value

34. Consider the decomposition of calcium sulfate:



$K_p = 0.032$ at 700 K. A sample of CaSO_4 sits in a sealed container at 700 K long enough to reach equilibrium. What is the total pressure in the container once equilibrium is reached? Assume no other gases were in the container at the start.

- A) 1.2 atm C) 0.60 atm E) 0.80 atm
B) 0.20 atm D) 0.40 atm

35. A tank holding 2500 liters of water has a pH of 3.50 due to the presence of HCl. How many grams of NaOH must be added to the tank so that the pH is brought to 4.34?

- A) 59 g C) 5 g E) 27 g
B) 32 g D) 38 g

36. What is the molecular geometry of XeF_4 ?

- A) square planar
B) see-saw
C) tetrahedral
D) square pyramid
E) bipyramidal

37. Which of the following neutral atoms will be the easiest to ionize into a +1 cation?

- A) F C) I E) Fe
B) Rb D) Sc

38. Acetone has the formula CH_3COCH_3 . What is the hybridization of the carbon in the middle of the formula?

- A) sp^3 C) sp^3d E) sp^2
B) sp^3d^2 D) sp

39. A first order reaction has $k = 3.2 \times 10^{-4} \text{ s}^{-1}$ at 300 K. If the reaction begins with 0.15 M of the reactant, how long will it take for the reactant to fall to 0.01M?

- A) 7.3 min D) 1.02 hr
B) 81.0 hr E) 9.13 hr
C) 2.35 hr

40. A chemical reaction where $\Delta H_{\text{rxn}} = +23.5 \text{ kJ/mol}$ has an equilibrium constant equal to 4.2×10^{-4} when the reaction is run at 100°C . What is the value of the equilibrium constant when the reaction is run at 300°C ?

- A) 4.2×10^{-4} D) 5.9×10^{-3}
B) 8.3×10^{-1} E) 6.4×10^4
C) 7.6×10^{-2}

Physics Questions (41 – 60)

On this contest $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors in the $+x$, $+y$ and $+z$ directions respectively, and unless otherwise indicated, you may neglect air resistance in every problem.

41. According to Tyson, what is Pluto's largest moon?
A) Charon
B) Europa
C) Hydra
D) Nix
E) Phobos
42. According to Tyson, what geometrical shape has the least surface area but the greatest enclosed volume?
A) cube
B) cylinder
C) ellipsoid
D) pyramid
E) sphere
43. According to Tyson, what spacecraft was launched with the stated goal to "complete the reconnaissance of the solar system"?
A) Hubble Space Telescope
B) New Horizons
C) Pioneer 10
D) Pioneer 11
E) Voyager
44. According to Tyson, which of the following is not included in the "Pluto system"?
A) Charon
B) Hydra
C) Nix
D) Phobos
E) Pluto
45. Given the following two vectors in unit vector notation:
 $\mathbf{A} = (3.00 \text{ m}) \mathbf{i} + (3.00 \text{ m}) \mathbf{j} + (5.00 \text{ m}) \mathbf{k}$
 $\mathbf{B} = (2.00 \text{ m}) \mathbf{i} - (3.00 \text{ m}) \mathbf{k}$
Determine $\mathbf{A} + \mathbf{B}$ in unit vector notation.
A) $(5.00 \text{ m}) \mathbf{i} + (5.00 \text{ m}) \mathbf{k}$
B) $(6.00 \text{ m}) \mathbf{i} - (15.00 \text{ m}) \mathbf{k}$
C) $(5.00 \text{ m}) \mathbf{i} + (3.00 \text{ m}) \mathbf{j} + (2.00 \text{ m}) \mathbf{k}$
D) $(5.00 \text{ m}) \mathbf{i} + (3.00 \text{ m}) \mathbf{j} + (8.00 \text{ m}) \mathbf{k}$
E) $(5.00 \text{ m}) \mathbf{i} + (6.00 \text{ m}) \mathbf{j} + (5.00 \text{ m}) \mathbf{k}$
46. A small fishing boat moves with a constant velocity of 12.0 m/s for 5 minutes. Then the fisherman stops the motor and lets the boat coast while fishing. If the coasting velocity is given by $v_c(t) = v(t_1) \times [t_1/t]^2$ where $t_1 = 5$ minutes. What is the boats displacement from $t_1 < t < \infty$?
A) 1200 m
B) 2400 m
C) 3600 m
D) 7200 m
E) 10800 m
47. While standing still on the sidewalk you watch a taxi move away from you with a constant velocity. Which of the following is true?
A) gravity is the only force acting on the taxi
B) the net force acting on the taxi is in the opposite direction of motion
C) the net force acting on the taxi in is the direction of motion
D) net force on the taxi is zero
E) no forces are acting on the taxi
48. In the backyard a 25.0 kg little girl swings on a 7.00 m long swing with a maximum speed of 5.60 m/s. What is the vertical angle that the swing makes at when the girl is at the highest point in her motion? You may neglect the mass of the swing and treat the girl as a point mass on the end of the 7.00 m long swing.
A) 16.4°
B) 39.5°
C) 57.1°
D) 62.8°
E) 76.8°

49. A tangential force is applied at a distance d from the fixed axis of a freely rotating disk. This force causes an angular acceleration of α to occur. If the same force is applied at a distance $4d$ from the axis then what is the new angular acceleration?
- $(1/4)\alpha$
 - $(1/2)\alpha$
 - α
 - 2α
 - 4α
50. On the moon's surface gravity is given by g_m . A point is located at a distance three times the radius of the moon from the moon's center. What is the moon's acceleration due gravity at this point?
- $(1/9)g_m$
 - $(1/3)g_m$
 - g_m
 - $3g_m$
 - $9g_m$
51. A 13.0 kg box is hung from a steel wire with a diameter of 0.800 mm and an initial length of 1.60 m. Given that Young's modulus for steel is 200 GPa, then how much will the wire stretch under this load?
- 8.11×10^{-7} m
 - 4.06×10^{-7} m
 - 5.07×10^{-5} m
 - 6.45×10^{-4} m
 - 2.03×10^{-3} m
52. An object attached to a spring oscillates on a horizontal frictionless surface with an amplitude of 5.70 cm. If the system has a total mechanical energy of 1.70 J, then what is the spring constant (or force constant) of the spring?
- 29.8 N/m
 - 59.7 N/m
 - 262 N/m
 - 523 N/m
 - 1050 N/m
53. To increase the sound intensity level by 20 dB means you need to increase the sound intensity by a factor of
- 2.0
 - 3.2
 - 10
 - 20
 - 100
54. If a gas absorbs 2.25 MJ of heat when doing 950 kJ of work then what is the change of the gas's internal energy?
- 2.25×10^3 J
 - 9.50×10^5 J
 - 1.30×10^6 J
 - 2.25×10^6 J
 - 3.20×10^6 J
55. How much energy is required to accelerate a particle of mass m from rest to $0.500c$?
- $0.155mc^2$
 - $0.330mc^2$
 - $0.577mc^2$
 - $1.15mc^2$
 - $3.00mc^2$
56. If a positively charged particle is allowed to freely move within a region of electric field E then which of the following is always true
- the particle is moving in the direction of E
 - the particle can come to rest and stay at rest
 - the particle accelerates in a direction perpendicular to the direction of E
 - the particle experiences an electric force perpendicular to the direction of E
 - the particle experiences an electric force in the direction of E

57. The electric potential at the surface of a uniformly charged sphere is 620 V. At a point located 30.0 cm radially from the surface of the sphere the electric potential is 230 V. Given that at a great distance from the sphere the electric potential is zero, then what is the radius of the sphere?
- A) 0.081 m
 - B) 0.088 m
 - C) 0.177 m
 - D) 0.219 m
 - E) 0.481 m
58. The electric field measured at the surface of the Earth is 150 N/C pointing downward. If this is the value over the entire surface and modeling the Earth as a sphere, then what is the net charge of the Earth?
- A) $-2.71 \times 10^6 \text{ C}$
 - B) $-6.77 \times 10^5 \text{ C}$
 - C) $+2.71 \times 10^5 \text{ C}$
 - D) $+6.77 \times 10^5 \text{ C}$
 - E) $+2.71 \times 10^6 \text{ C}$
59. A series RLC circuit combo with a resistance of $15.0 \, \Omega$, an inductance of 4.00 H, and a capacitance of $3.00 \, \mu\text{F}$ is driven by an ideal generator with a peak emf of 125 V. What is the resonance frequency?
- A) 45.9 Hz
 - B) 115 Hz
 - C) 289 Hz
 - D) 1810 Hz
 - E) 13300 Hz
60. A goldfish is 8.0 cm away from the front glass in a spherical bowl with a radius of 25 cm. How far behind the glass does the goldfish appear to be to a woman looking through the front of the bowl? You may neglect the effects of the glass bowl.
- A) 5.6 cm
 - B) 6.0 cm
 - C) 6.5 cm
 - D) 70 cm
 - E) 76 cm

UIL HIGH SCHOOL SCIENCE CONTEST

Contestant Answer Sheet

Contestant # _____

9 10 11 12
CONTESTANT GRADE LEVEL

A 2A 3A 4A 5A
CONFERENCE

PLEASE WRITE ANSWERS WITH CAPITAL LETTERS

- | | | |
|-----------|-----------|-----------|
| 1. _____ | 21. _____ | 41. _____ |
| 2. _____ | 22. _____ | 42. _____ |
| 3. _____ | 23. _____ | 43. _____ |
| 4. _____ | 24. _____ | 44. _____ |
| 5. _____ | 25. _____ | 45. _____ |
| 6. _____ | 26. _____ | 46. _____ |
| 7. _____ | 27. _____ | 47. _____ |
| 8. _____ | 28. _____ | 48. _____ |
| 9. _____ | 29. _____ | 49. _____ |
| 10. _____ | 30. _____ | 50. _____ |
| 11. _____ | 31. _____ | 51. _____ |
| 12. _____ | 32. _____ | 52. _____ |
| 13. _____ | 33. _____ | 53. _____ |
| 14. _____ | 34. _____ | 54. _____ |
| 15. _____ | 35. _____ | 55. _____ |
| 16. _____ | 36. _____ | 56. _____ |
| 17. _____ | 37. _____ | 57. _____ |
| 18. _____ | 38. _____ | 58. _____ |
| 19. _____ | 39. _____ | 59. _____ |
| 20. _____ | 40. _____ | 60. _____ |

BIOLOGY SCORE

CHEMISTRY SCORE

PHYSICS SCORE

OVERALL SCORE:

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY**

DISTRICT 1 • 2015

- | | | |
|-------|------------------|-------|
| 1. D | 21. E | 41. A |
| 2. A | 22. C | 42. E |
| 3. D | 23. A | 43. B |
| 4. D | 24. D | 44. D |
| 5. B | 25. D | 45. C |
| 6. C | 26. A | 46. C |
| 7. E | 27. C | 47. D |
| 8. B | 28. B | 48. B |
| 9. D | 29. E | 49. E |
| 10. D | 30. A | 50. A |
| 11. B | 31. B | 51. E |
| 12. B | 32. C | 52. E |
| 13. D | 33. D | 53. E |
| 14. A | 34. C | 54. C |
| 15. C | 35. E | 55. A |
| 16. C | 36. A | 56. E |
| 17. D | 37. B | 57. C |
| 18. B | 38. E | 58. B |
| 19. C | 39. C | 59. A |
| 20. D | 40. D | 60. C |

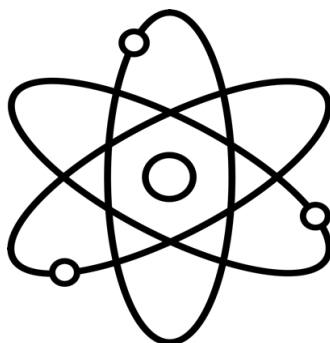
PHYSICS KEY for Science Contest • District 1 • 2015

41. (A) "Pluto's largest moon, Charon, named for the Greek ferryboat driver who would carry your unfortunate soul across the River Acheron to the underworld, ..."
42. (E) "You can show, using freshman-level calculus, that the one and only shape that has the smallest surface area for an enclosed volume is a perfect sphere."
43. (B) "One of the stated goals for the New Horizons mission was to "complete the reconnaissance of the solar system.""
44. (D) "Two famous celestial nonspheres are Phobos and Deimos, the Idaho-potato-shaped moons of Mars."
45. (C) $\mathbf{A} + \mathbf{B} = [(3.00 \text{ m}) \mathbf{i} + (3.00 \text{ m}) \mathbf{j} + (5.00 \text{ m}) \mathbf{k}] + [(2.00 \text{ m}) \mathbf{i} + (0.00 \text{ m}) \mathbf{j} - (3.00 \text{ m}) \mathbf{k}]$
 $= (5.00 \text{ m}) \mathbf{i} + (3.00 \text{ m}) \mathbf{j} + (2.00 \text{ m}) \mathbf{k}$
46. (C) $\Delta x = \int v_c(t) dt = \int v(t_1) \times [t_1/t]^2 dt = v(t_1) \times t_1^2 \int [1/t]^2 dt = v(t_1) \times t_1^2 (-1/t|_{t_1}^{\infty}) = v(t_1) \times t_1 = 12[5(60)] = 3600 \text{ m}$
47. (D) Since the taxi moves with a constant velocity then the acceleration of the taxi is zero, and thus the net force on the taxi is zero.
48. (B) By conservation of mechanical energy: $\frac{1}{2}mv^2 = mgh \Rightarrow h = v^2/(2g) = 1.6 \text{ m}$, then by the diagram of the situation $\theta = \cos^{-1}([7.0 - 1.6]/7.0) = 39.5^\circ$
49. (E) By N2L in angular form: $\alpha = \Sigma \tau/I = (Fd)/I$, thus if $d \rightarrow 4d$ then $\alpha \rightarrow 4\alpha$
50. (A) With $g_m = GM_m/R_m^2$, then $g = GM_m/r^2$ with $r = 3R_m \Rightarrow g = GM_m/(3R_m)^2 = (1/9)g_m$
51. (E) By $F/A = Y\Delta L/L_0 \Rightarrow \Delta L = [FL_0]/[YA] = [(13)(9.8)(1.6)]/[(2E+11)\pi(0.4E-3)^2] = 2.03E-3 \text{ m}$
52. (E) By $E_{\text{sho}} = \frac{1}{2}kA^2 \Rightarrow k = 2E_{\text{sho}}/A^2 = 2(1.7)/(0.057)^2 = 1050 \text{ N/m}$
53. (E) By $\beta = 10 \log(I/I_0) \Rightarrow \beta' = \beta + 20 = 10 \log(I'/I_0) \Rightarrow 20 = 10[\log(I'/I_0) - \log(I/I_0)] = 10 \log(I'/I)$
 $\Rightarrow I'/I = 10^2 = 100$
54. (C) $\Delta E_{\text{int}} = Q - W = (+2.25E+6) - (+950E+3) = 1.30E+6 \text{ J}$
55. (A) $\Delta E_{\text{rel}} = \gamma mc^2 - \gamma_0 mc^2 = [1 - 0.5^2]^{-1/2} mc^2 - mc^2 = 0.155mc^2$
56. (E) The electric force on a positively charged particle is always in the same direction as the electric field vector.
57. (C) With $V = kQ/r \Rightarrow 620 = kQ/R$ & $230 = kQ/(R+0.3) \Rightarrow R = [(230)(0.3)]/[620-230] = 0.177 \text{ m}$
58. (B) By Gauss' law: $\Phi_E = \oint \mathbf{E} \cdot d\mathbf{a} = Q_{\text{enc}}/\epsilon_0 = -EA \Rightarrow Q_{\text{enc}} = -(8.85E-12)(150)[4\pi(6.37E+6)^2] = -6.77E+5 \text{ C}$
59. (A) With $f_0 = 1/\{2\pi[LC]^{1/2}\} = 1/\{2\pi[(4)(3E-6)]^{1/2}\} = 45.9 \text{ Hz}$
60. (C) By $n_1/d_o + n_2/d_i = (n_2 - n_1)/r \Rightarrow (1.33/8) + (1.00/d_i) = (1.00 - 1.33)/(-25) \Rightarrow d_i = -6.5 \text{ cm}$



SCIENCE

District 2 • 2015



GENERAL DIRECTIONS:

- DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
- Ninety minutes should be ample time to complete this contest, but since it is not a race, contestants may take up to two hours. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet.
- You may place as many notations as you desire anywhere on the test paper except on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- On the back of this page is printed a copy of the periodic table of the elements. You may wish to refer to this table in answering the questions, and if needed, you may use the atomic weights and atomic numbers from the table. Other scientific relationships are listed also.
- Silent hand-held calculators that do not need external wall plugs may be used. Graphing calculators that do not have built-in or stored functionality that provides additional scientific information are allowed. Small hand-held computers are not permitted. Calculators that accept memory cards or memory sticks are not permitted. Each contestant may bring one spare calculator. All memory must be cleared.

SCORING:

All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for an incorrect answer.

Periodic Table of the Elements

1A 1		2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.008																		2 He 4.003
3 Li 6.941	4 Be 9.012												5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12		13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41		31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41		49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59		81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (285)							

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Some Standard Properties of Water

property	symbol	value
density of water	ρ_{water}	1.000 g cm ⁻³
density of ice	ρ_{ice}	0.9167 g cm ⁻³
specific heats		
ice	C_{ice}	2.09 J g ⁻¹ K ⁻¹
water	C_{water}	4.184 J g ⁻¹ K ⁻¹
steam	C_{steam}	2.03 J g ⁻¹ K ⁻¹
heat of fusion	ΔH_{fus} or L_f	334 J g ⁻¹
heat of vaporization	ΔH_{vap} or L_v	2260 J g ⁻¹
index of refraction	n	1.33
autoionization	K_w	1.0×10^{-14}

Pressure		
1 atm	=	760 torr
	=	101325 Pa
	=	14.7 psi
1 bar	=	10 ⁵ Pa
	=	100 kPa

Energy		
1 cal	=	4.184 J
1 L atm	=	101.325 J
1 Cal	=	1 kcal
1 hp	=	746 W
1 eV	=	1.602×10^{-19} J

Various Physical Constants

property	symbol	value
universal gas constant	R	8.314 J mol ⁻¹ K ⁻¹ 62.36 L torr mol ⁻¹ K ⁻¹ 0.08206 L atm mol ⁻¹ K ⁻¹ 1.987 cal mol ⁻¹ K ⁻¹
Planck's constant	h	6.626×10^{-34} J s 4.136×10^{-15} eV s
Planck's reduced constant	$\hbar/2\pi$	1.054×10^{-34} J s 6.582×10^{-16} eV s
Boltzmann constant	k_B	1.38×10^{-23} J K ⁻¹
Stefan-Boltzmann	σ	5.67×10^{-8} W m ⁻² K ⁻⁴
speed of light	c	3.00×10^8 m s ⁻¹
speed of sound (at 20°C)	v_{air}	343 m s ⁻¹
acceleration of gravity	g	9.80 m s ⁻²
gravitational constant	G	6.67×10^{-11} N m ² kg ⁻²
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
elementary charge	e	1.602×10^{-19} C
Faraday	F	96485 C mol ⁻¹
Coulomb's law constant	k	8.988×10^9 N m ² C ⁻²
Rydberg constant	R	2.178×10^{-18} J

Some Other Conversion Factors

1 in	=	2.54 cm
1 lb	=	453.6 g
1 mi	=	5280 ft = 1.609 km
1 gal	=	4 quarts = 231 in ³ = 3.785 L

property	symbol	value
electron rest mass	m_e	9.11×10^{-31} kg 0.000549 u 0.511 MeV c ⁻²
proton mass	m_p	1.6726×10^{-27} kg 1.00728 u 938.3 MeV c ⁻²
neutron mass	m_n	1.6749×10^{-27} kg 1.008665 u 939.6 MeV c ⁻²
atomic mass unit	u	1.6605×10^{-27} kg 931.5 MeV c ⁻²
earth mass		5.972×10^{24} kg
earth radius		6.371×10^6 m
moon mass		7.348×10^{22} kg
sun mass		1.989×10^{30} kg
distance earth-moon		3.844×10^8 m
distance earth-sun		1.496×10^{11} m
permittivity of free space	ϵ_0	8.85×10^{-12} F m ⁻¹
permeability of free space	μ_0	$4\pi \times 10^{-7}$ T m A ⁻¹

Some Average Bond Energies (kJ/mol)

C-H	413	C-C	346	C-Cl	339	C-N	305
O-H	463	C=C	602	C-Br	285	N=N	418
N-H	391	C≡C	835	O=O	498	H-H	436
C-O	358	C=O	799	C≡O	1072	Br-Br	193
H-Cl	432	S-H	347	N≡N	945	Cl-Cl	242
H-Br	366	H-I	299	C≡N	887	I-I	151

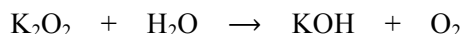
Biology Questions (1 – 20)

1. In eukaryotic cells, glycolysis occurs
 - A) In the same cellular location as in a prokaryotic cell
 - B) In a different cellular location than in a prokaryotic cell.
 - C) Glycolysis does not occur in eukaryotic cells.
2. Genetic engineering relies on the use of _____ which are enzymes that cut DNA at a limited number of specific locations.
 - A) restriction endonucleases
 - B) plasmid vectors
 - C) recombinant DNAses
 - D) DNA ligases
 - E) RNA ligases
3. What is the complementary RNA sequence for a segment of DNA with the sequence: 5' TTA GCG 3'
 - A) 5' AAT CGC 3'
 - B) 3' AAT CGC 5'
 - C) 5' AAU CGC 3'
 - D) 3' AAU CGC 5'
 - E) 5' CGC TAA 3'
4. Which of the following is true when a metabolic reaction is at equilibrium?
 - A) consumption of products is at a maximum
 - B) synthesis of reactants is at a minimum
 - C) no net release of free energy can occur
 - D) the reaction can power an endergonic reaction
 - E) the effect of inhibitors can be seen
5. Bryophytes are non-vascular land plants. Which trait do they all share?
 - A) use diffusion to move water and nutrients through the plant.
 - B) release diploid spores
 - C) are woody perennials
 - D) have a fibrous root system
 - E) lack mitochondria
6. Natural selection requires three basic conditions to occur. Which of the following is NOT required for a population to evolve by natural selection?
 - A) random mating
 - B) differential reproduction
 - C) heritable variation
7. Common characteristics of arthropods do not include which of the following?
 - A) Segmented bodies structure
 - B) Exoskeleton made of rigid protein called chitin
 - C) Jointed appendages
 - D) Body made of a head, abdomen, and thorax
8. Which of the following are the most abundant organisms by number on Earth?
 - A) insects
 - B) angiosperms
 - C) microbes
 - D) fungi
9. Which of the following terms are mismatched?
 - A) Spherical bacteria: Cocci
 - B) Rod-shaped bacteria: Linnelli
 - C) Spiral-shaped bacteria: Spirilli
10. Carbon is found in four main compartments on the Earth: the oceans, the atmosphere, fossil deposits and _____.
 - A) phytoplankton
 - B) thermal vents
 - C) terrestrial organisms
 - D) rock strata
11. The presence of a _____ species has a strong positive effect on community diversity.
 - A) target
 - B) keystone
 - C) clutch
 - D) dominant
 - E) diverse

12. Alternating waves of contraction and relaxation in the smooth muscle lining the alimentary canal is called:
A) secretion
B) peristalsis
C) excretion
D) osmostasis
E) interstitis
13. In animals, sheets of cells that cover the body and line organs and cavities are known as _____ tissue.
A) connective
B) squamous
C) columnar
D) adipose
E) epithelial
14. Ventilation is the movement of air across a thin respiratory surface that promotes the diffusion of gases. Which of the following can function as a respiratory surface?
A) tracheal systems
B) gills
C) skin
D) B and C
E) A, B and C
15. The transmission of traits from one generation to another may require which of the following?
A) mitosis
B) meiosis
C) fertilization
D) B and C
E) A, B and C
16. Mendel's law of independent assortment can only be observed in:
A) inheritance of alleles at the same locus
B) inheritance of alleles at two different loci on two different chromosomes
C) inheritance of alleles at two different loci on the same chromosome
D) inheritance of parental phenotypes
E) inheritance of homozygous alleles
17. The organelle that receives transport vesicles from the rough ER and is made up of flattened stacks of cisternae that differ in thickness and molecular composition is the:
A) Golgi apparatus
B) Ribosome
C) Mitochondria
D) Smooth ER
E) ER lumen
18. The act of fusion of a vesicle containing a damaged organelle with a lysosome is known as:
A) coprophagy
B) necrophagy
C) autophagy
D) phagocytosis
E) pinocytosis
19. The change in free energy of a system (ΔG) can be calculated as the difference between the change in the system's _____ and the change in the system's _____ at a given temperature (in K).
A) energy; heat
B) enthalpy; entropy
C) entropy; enthalpy
D) entropy; heat
E) energy; enthalpy
20. The range of phenotypes observed for a particular genotype across a range of environment is known as?
A) dominance
B) epistasis
C) norm of reaction
D) normal distribution
E) Chi-square distribution
21. What is the formula for copper(II) chromate?
A) $\text{Cu}(\text{CrO}_4)_2$
B) CoCr_2O_7
C) CuCrO_4
D) CoCO_3
E) $\text{Cu}_2\text{Cr}_2\text{O}_7$

Chemistry Questions (21 – 40)

22. Balance the following reaction with whole numbers as coefficients.



What are the respective coefficients?

- A) 1 : 1 : 2 : 1
 B) 2 : 1 : 4 : 2
 C) 3 : 3 : 6 : 2
 D) 1 : 2 : 2 : 1
 E) 2 : 2 : 4 : 1
23. A laboratory bottle is labeled 3.75 M H_2SO_4 with a density of 1.251 g/mL. What is the molality of the H_2SO_4 ?

- A) 4.25 *m*
 B) 3.75 *m*
 C) 4.15 *m*
 D) 3.95 *m*
 E) 4.45 *m*

24. The solvent carbon tetrachloride is classified as _____.

- A) an ionic compound
 B) a nonpolar covalent compound
 C) a polar covalent compound
 D) an resonance compound
 E) a covalent metalloid compound

25. Write the proper line/dot structure for the sulfate ion. What is the formal charge on the sulfur atom in this ion?

- A) +2 C) -1 E) 0
 B) -2 D) +1

26. 355 g of chlorine gas is added to a 50-L evacuated container at 28°C. What is the pressure of this chlorine gas?

- A) 72.8 psi C) 23.9 psi E) 51.2 psi
 B) 3.38 psi D) 36.4 psi

27. The value for $\Delta H_{\text{solution}}$ under aqueous conditions is best thought of (and calculated) as a combination of what other two enthalpy changes?

- A) $\Delta H_{\text{hydration}}$ and ΔH_{fusion}
 B) $\Delta H_{\text{lattice}}$ and $\Delta H_{\text{hydration}}$
 C) $\Delta H_{\text{formation}}$ and $\Delta H_{\text{hydration}}$
 D) $\Delta H_{\text{lattice}}$ and $\Delta H_{\text{formation}}$
 E) $\Delta H_{\text{combustion}}$ and $\Delta H_{\text{hydration}}$

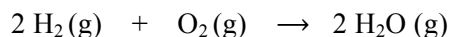
28. Lead(II) iodate has a K_{sp} value equal to 2.6×10^{-13} . A saturated solution of lead(II) iodate has an iodate concentration of 1.6×10^{-4} M. What is the concentration of lead(II) in this solution?

- A) 2.5×10^{-6} M
 B) 1.6×10^{-9} M
 C) 4.0×10^{-5} M
 D) 6.2×10^{-7} M
 E) 1.0×10^{-5} M

29. Which solution listed has the lowest vapor pressure?

- A) 1.7 *m* NaCl D) 1.2 *m* $\text{Sr}(\text{NO}_3)_2$
 B) 1.0 *m* CaCl_2 E) 0.50 *m* $\text{Al}(\text{NO}_3)_3$
 C) 0.80 *m* Na_3PO_4

30. Consider a system at 300 K with the following gas phase reaction occurring:



What is the amount of work done? Is the work done *on* the system or *by* the system?

- A) 7.5 kJ of work done on the system
 B) 2.5 kJ of work done by the system
 C) no work done on or by the system
 D) 7.5 kJ of work done by the system
 E) 2.5 kJ of work done on the system

31. Chloroform (CHCl_3) has the following phase data:

$$\begin{aligned} \Delta H_{\text{fusion}} &= 8.80 \text{ kJ/mol} & T_{\text{fp}} &= -63^\circ\text{C} \\ \Delta H_{\text{vaporization}} &= 30.8 \text{ kJ/mol} & T_{\text{bp}} &= 61^\circ\text{C} \\ C_{\text{p,liquid}} &= 114. \text{ J/mol K} \end{aligned}$$

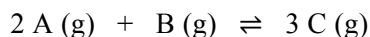
How much heat must be added to 185 g of chloroform at 25°C in order to change all of it to gas state at 61°C?

- A) 69.7 kJ C) 74.2 kJ E) 61.0 kJ
 B) 54.1 kJ D) 20.0 kJ

32. Which example has $\Delta S < 0$ for the change given?

- A) a dash of salt dissolves into water
- B) a cube of dry ice sublimates
- C) a gas molecule splits into two other gas molecules
- D) water vapor condenses on a cold windshield
- E) an aluminum block is warmed from 20°C to 23°C

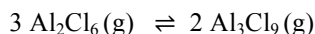
33. Consider the gas phase equilibrium:



If the partial pressure of A is doubled, what happens to the reaction quotient, Q ?

- A) it drops to $1/4^{\text{th}}$ its original value
- B) it increases by a factor of 2
- C) Q will not change
- D) it increases by a factor of 4
- E) it drops to $1/2^{\text{th}}$ its original value

34. A sample of Al_2Cl_6 is put into a gas reaction chamber at 500K and allowed to equilibrate according to the reaction shown:



At equilibrium the partial pressure of Al_3Cl_9 is 5.0×10^{-3} atm and the Al_2Cl_6 is 1.1 atm. What is the value of the equilibrium constant, K ?

- A) 1.0×10^{-7}
- B) 4.5×10^{-3}
- C) 1.9×10^{-5}
- D) 2.5×10^{-5}
- E) 2.1×10^{-5}

35. An Olympic sized pool maintained at 25°C has one million gallons of water currently at a pH of 7.00. One pound of NaOH is added to the pool. Assuming that there is no buffering, approximate the new pH after all the NaOH has dissolved and mixed?

- A) 8.48
- B) 7.56
- C) 5.37
- D) 8.63
- E) 9.14

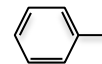
36. What is approximate bond angle ($\angle \text{FIF}$) in the ion, IF_5^- ?

- A) 75°
- B) 60°
- C) 109.5°
- D) 90°
- E) 120°

37. Which of the following neutral atoms has the largest electron affinity?

- A) Be
- B) Br
- C) He
- D) Rb
- E) O

38. Consider the line structure for toluene shown. How many sp^3 hybridized atoms are there in this structure?



- A) one
- B) two
- C) three
- D) six
- E) seven

39. A first order reaction takes 3 hours to fall from a concentration of 0.25 M to 0.015 M at a temperature of 310 K. What is the value of the rate constant, k ?

- A) $5.8 \times 10^{-3} \text{ s}^{-1}$
- B) $1.1 \times 10^{-4} \text{ s}^{-1}$
- C) $2.6 \times 10^{-4} \text{ s}^{-1}$
- D) $6.8 \times 10^{-5} \text{ s}^{-1}$
- E) $9.4 \times 10^{-1} \text{ s}^{-1}$

40. The equilibrium constant for a reaction is doubled when the temperature is changed from 25°C to 100°C. What is the value of ΔH for this reaction?

- A) -9.25 kJ
- B) $+8.55 \text{ kJ}$
- C) $+3.71 \text{ kJ}$
- D) $-341. \text{ J}$
- E) $+192. \text{ J}$

Physics Questions (41 – 60)

On this contest $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors in the $+x$, $+y$ and $+z$ directions respectively, and unless otherwise indicated, you may neglect air resistance in every problem.

41. According to Tyson, which asteroid was discovered to have a moon placing doubt on the idea that only planets can have moons?
A) Ceres
B) Dactyl
C) Hydra
D) Ida
E) Vesta
42. According to Tyson, what planetary astronomer suggested that “beyond the outermost planet in the solar system lies a reservoir of slowly orbiting debris”?
A) Ann Harch
B) David Jewitt
C) Gerard Kuiper
D) Jane Luu
E) Clyde Tombaugh
43. According to Tyson, which of the following expert scientist did NOT appear onstage with the author, Neil DeGrasse Tyson, to debate the classification of Pluto?
A) Johannes Anderson
B) Michael A’Hearn
C) David H. Levy
D) Jane Luu
E) Brian Marsden
44. According to Tyson, how did the Rose Center’s Cullman Hall of the Universe deal with the conundrum of Pluto’s placement in the universe?
A) declared that the universe only has eight planets
B) listed Pluto as an asteroid
C) listed Pluto as a comet in the Oort cloud
D) listed Pluto as a planet
E) used a familial treatment of the solar system
45. Which of the following is NOT a base unit in the SI system?
A) ampere
B) gram
C) meter
D) mole
E) seconds
46. A particle has a position vector given by $\mathbf{r}(t) = [(50t)\mathbf{i} + (20t - 12t^2)\mathbf{j}]$ m. What is the instantaneous velocity vector?
A) $[10(29)^{1/2}]$ m/s
B) $[50\mathbf{i} + 20\mathbf{j}]$ m/s
C) $[-(24t)\mathbf{j}]$ m/s
D) $[50\mathbf{i} + (20 - 24t)\mathbf{j}]$ m/s
E) $[(25t^2)\mathbf{i} + (10t^2 - 4.0t^3)\mathbf{j}]$ m/s
47. Imagine that a car attached to the rim of a Ferris wheel undergoes uniform circular motion as it is rotated in a vertical circle. Which of the following is NOT a constant quantity for the car in this situation?
A) magnitude of the acceleration of the car
B) magnitude of the force of gravity on the car
C) magnitude of the force on the car by the rim
D) magnitude of the net force on the car
E) magnitude of velocity of the car
48. A 110 kg man runs off of a horizontal river bank to land on a freely floating raft with mass of 175 kg that is initially at rest. After impact, the raft/man combo moves away from the bank with a horizontal speed of 3.50 m/s. What was the horizontal speed of the man as he left the river bank?
A) 1.59 m/s
B) 2.07 m/s
C) 2.59 m/s
D) 5.57 m/s
E) 9.07 m/s

49. A 5.25 kg uniform cylinder with a radius of 12.0 cm and a height of 15.0 cm can freely rotate about an axis that runs along the symmetry axis passing through the center of mass. If an ideal string that is wrapped around the cylinder is used to apply a constant tangential force of 25.0 N, then what is the angular acceleration of the cylinder? You may assume that the string does not slip.
- 1.14 rad/s^2
 - 1.59 rad/s^2
 - 9.52 rad/s^2
 - 39.7 rad/s^2
 - 79.4 rad/s^2
50. A 4.50 kg particle moves due east with a constant speed of 5.00 m/s. What is the angular momentum of the particle about a point 7.50 m due north of the path of motion?
- $0.00 \text{ kg}\cdot\text{m}^2/\text{s}$
 - $119 \text{ kg}\cdot\text{m}^2/\text{s}$, upward
 - $119 \text{ kg}\cdot\text{m}^2/\text{s}$, downward
 - $169 \text{ kg}\cdot\text{m}^2/\text{s}$, upward
 - $169 \text{ kg}\cdot\text{m}^2/\text{s}$, downward
51. A planet orbits a massive star in an elliptical orbit such that at perihelion the planet is $1.5 \times 10^{15} \text{ m}$ from the star and at aphelion the planet is $2.7 \times 10^{15} \text{ m}$ away from the star. If the planet has an orbital speed of $7.3 \times 10^4 \text{ m/s}$ at perihelion, then what is the orbital speed of the planet at aphelion?
- $2.3 \times 10^4 \text{ m/s}$
 - $4.1 \times 10^4 \text{ m/s}$
 - $7.3 \times 10^4 \text{ m/s}$
 - $1.3 \times 10^5 \text{ m/s}$
 - $4.7 \times 10^5 \text{ m/s}$
52. A horizontal pipe narrows from 20 cm in diameter at point A to 10 cm in diameter at point B. How do the flow speeds at the two points compare if the pipe is filled with nonviscous, incompressible fluid with no turbulence?
- $v_A = (1/4)v_B$
 - $v_A = (1/2)v_B$
 - $v_A = v_B$
 - $v_A = 2v_B$
 - $v_A = 4v_B$
53. Suppose you have two mass-spring systems, A and B, which oscillate such that they have the same total mechanical energies. If the spring (or force) constant of A is twice that of B, then how does the amplitude of A, A_A , relate to the amplitude of B, A_B ?
- $A_A = A_B/2$
 - $A_A = A_B/(2)^{1/2}$
 - $A_A = A_B$
 - $A_A = (2)^{1/2}A_B$
 - $A_A = 2A_B$
54. A 5.50 m long string with a mass of 0.250 kg is fixed at both ends and has a tension of 50.0 N. What is the frequency of the 1st harmonic for transverse waves on this string?
- 3.02 Hz
 - 4.26 Hz
 - 6.03 Hz
 - 100 Hz
 - 365 Hz
55. A uniform line of charge with a constant linear charge density of $+6.5 \text{ nC/m}$ is on the x -axis between 0.0 and 10.0 m. Find the magnitude of the electric field on the x -axis at $x = 12.0 \text{ m}$.
- 24 N/C
 - 47 N/C
 - 97 N/C
 - 290 N/C
 - 580 N/C
56. A point particle has a $+5.0 \mu\text{C}$ charge and is fixed at the origin. How much work is done by an external agent to bring a 2nd point particle with a charge of $+4.0 \mu\text{C}$ from infinity to a distance of 6.0 m from the 1st charge?
- $+1.5 \times 10^{-7} \text{ J}$
 - $+2.0 \times 10^{-4} \text{ J}$
 - $+1.2 \times 10^{-3} \text{ J}$
 - $+3.0 \times 10^{-2} \text{ J}$
 - $+3.6 \times 10^4 \text{ J}$

57. A 20.0 μF capacitor and 25.0 μF capacitor are wired in series and this series combination is then connected across the terminals of a 12.0 V battery. What is the charge on each capacitor?
- A) $Q_{20} = 1.08 \mu\text{C}$ $Q_{25} = 1.08 \mu\text{C}$
 B) $Q_{20} = 133 \mu\text{C}$ $Q_{25} = 133 \mu\text{C}$
 C) $Q_{20} = 240 \mu\text{C}$ $Q_{25} = 300 \mu\text{C}$
 D) $Q_{20} = 500 \mu\text{C}$ $Q_{25} = 500 \mu\text{C}$
 E) $Q_{20} = 900 \mu\text{C}$ $Q_{25} = 1130 \mu\text{C}$
58. What is the magnetic force on an electron that has a velocity of $(3.0 \times 10^6 \mathbf{i} + 2.0 \times 10^6 \mathbf{j})$ m/s as it travels through a region with a uniform magnetic field given by $(0.65 \mathbf{i} - 0.45 \mathbf{j} + 0.35 \mathbf{k})$ T?
- A) $(-8.01 \times 10^{-15} \mathbf{i} + 1.12 \times 10^{-13} \mathbf{j} - 1.68 \times 10^{-13} \mathbf{k})$ N
 B) $(-1.12 \times 10^{-13} \mathbf{i} + 1.68 \times 10^{-13} \mathbf{j} + 8.01 \times 10^{-15} \mathbf{k})$ N
 C) $(1.12 \times 10^{-13} \mathbf{i} - 1.68 \times 10^{-13} \mathbf{j} - 8.01 \times 10^{-15} \mathbf{k})$ N
 D) $(-1.12 \times 10^{-13} \mathbf{i} + 1.68 \times 10^{-13} \mathbf{j} + 4.25 \times 10^{-13} \mathbf{k})$ N
 E) $(1.12 \times 10^{-13} \mathbf{i} - 1.68 \times 10^{-13} \mathbf{j} - 4.25 \times 10^{-13} \mathbf{k})$ N
59. A toroid with a radius of 30.0 cm is comprised of circular loops of radii 2.50 cm and is wound with a superconducting wire. If the wire has length of 1200 m and carries a current of 300 A, then what is the magnitude of the magnetic field strength at the center of the circular loops?
- A) 0.474 T
 B) 1.20 T
 C) 1.53 T
 D) 1.59 T
 E) 4.80 T
60. A 25 kW beam of electromagnetic radiation is normal to a surface that reflects half of the radiation. What force is exerted by the radiation on the surface?
- A) 8.3×10^{-8} N
 B) 4.1×10^{-8} N
 C) 2.5×10^{-7} N
 D) 1.3×10^{-4} N
 E) 7.5×10^4 N

UIL HIGH SCHOOL SCIENCE CONTEST

Contestant Answer Sheet

Contestant # _____

9 10 11 12
CONTESTANT GRADE LEVEL

A 2A 3A 4A 5A
CONFERENCE

PLEASE WRITE ANSWERS WITH CAPITAL LETTERS

- | | | |
|-----------|-----------|-----------|
| 1. _____ | 21. _____ | 41. _____ |
| 2. _____ | 22. _____ | 42. _____ |
| 3. _____ | 23. _____ | 43. _____ |
| 4. _____ | 24. _____ | 44. _____ |
| 5. _____ | 25. _____ | 45. _____ |
| 6. _____ | 26. _____ | 46. _____ |
| 7. _____ | 27. _____ | 47. _____ |
| 8. _____ | 28. _____ | 48. _____ |
| 9. _____ | 29. _____ | 49. _____ |
| 10. _____ | 30. _____ | 50. _____ |
| 11. _____ | 31. _____ | 51. _____ |
| 12. _____ | 32. _____ | 52. _____ |
| 13. _____ | 33. _____ | 53. _____ |
| 14. _____ | 34. _____ | 54. _____ |
| 15. _____ | 35. _____ | 55. _____ |
| 16. _____ | 36. _____ | 56. _____ |
| 17. _____ | 37. _____ | 57. _____ |
| 18. _____ | 38. _____ | 58. _____ |
| 19. _____ | 39. _____ | 59. _____ |
| 20. _____ | 40. _____ | 60. _____ |

BIOLOGY SCORE

CHEMISTRY SCORE

PHYSICS SCORE

OVERALL SCORE:

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY**

DISTRICT 2 • 2015

- | | | |
|-------|------------------|-------|
| 1. A | 21. C | 41. D |
| 2. A | 22. E | 42. C |
| 3. D | 23. A | 43. A |
| 4. C | 24. B | 44. E |
| 5. A | 25. E | 45. B |
| 6. A | 26. D | 46. D |
| 7. B | 27. B | 47. C |
| 8. C | 28. E | 48. E |
| 9. B | 29. D | 49. E |
| 10. C | 30. E | 50. D |
| 11. B | 31. B | 51. B |
| 12. B | 32. D | 52. A |
| 13. E | 33. A | 53. B |
| 14. E | 34. C | 54. A |
| 15. E | 35. A | 55. A |
| 16. B | 36. D | 56. D |
| 17. A | 37. E | 57. B |
| 18. C | 38. A | 58. D |
| 19. B | 39. C | 59. C |
| 20. C | 40. B | 60. D |

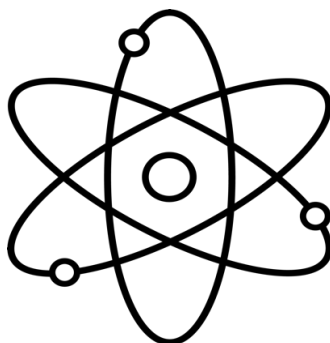
PHYSICS KEY for Science Contest • District 2 • 2015

41. (D) “While examining the data, mission member Ann Harch discovered that Ida has a small (1.4-kilometer) orbiting moon that came to be named Dactyl.”
42. (C) “Gerard Kupier had proposed that beyond the outermost planet in the solar system (perhaps in any star system) lies a reservoir of slowly orbiting debris...”
43. (A) “You couldn’t get more expert experts than the five scientists that joined me onstage:” Michael A’Hearn, David H. Levy, Jane Luu, Brian Marsden, and Alan Stern.
44. (E) “That’s when we decided to present the contents of the solar system as families of objects with similar properties...” “We knew that no matter how the Pluto debate would ultimately resolve, our familial treatment of the solar system was pedagogically and scientifically sensible – a kind of intellectual high road that sidestepped nomenclature altogether.”
45. (B) The SI base unit for mass is the kilogram.
46. (D) $\mathbf{v}(t) = (d/dt)[\mathbf{r}(t)] = [50\mathbf{i} + (20 - 24t)\mathbf{j}] \text{ m/s}$
47. (C) As the car undergoes vertical, uniform circular motion the magnitude of the force on the car by the rim will not be constant as can be seen by analyzing Newton’s second law at various points along the motion.
48. (E) By conservation of linear momentum: $110v_0 + 0 = (110+175)(+3.5) \Rightarrow v_0 = 9.07 \text{ m/s}$
49. (E) By $\alpha = \Sigma \tau/I = (rF)/(1/2mr^2) = (2F)/(mr) = [2(25)]/[(5.25)(0.12)] = 79.4 \text{ rad/s}^2$
50. (D) With $\mathbf{L} = \mathbf{r} \times \mathbf{p} \Rightarrow L = rmv \sin\phi = (7.5)(4.5)(5) \sin 90^\circ = 169 \text{ kg}\cdot\text{m}^2/\text{s}$ & by the right hand rule the direction is upward
51. (B) By conservation of angular momentum: $L_a = L_p \Rightarrow r_a m v_a \sin\phi_a = r_p m v_p \sin\phi_p \Rightarrow v_a = (r_p/r_a)v_p = [(1.5\text{E}+15)/(2.7\text{E}+15)](7.3\text{E}+4) = 4.1\text{E}+4 \text{ m/s}$
52. (A) For an incompressible fluid the volume flow rate, $Q = vA \cos\theta$, remains constant at any point along a pipe, thus $Q_A = Q_B \Rightarrow v_A (\pi/4)D_A^2 = v_B (\pi/4)D_B^2$ & with $D_B = 1/2 D_A \Rightarrow v_A = (1/4)v_B$
53. (B) By $E_{\text{sho}} = 1/2 k A^2 \Rightarrow 1/2 k_A A_A^2 = 1/2 k_B A_B^2$ & with $k_A = 2k_B \Rightarrow A_A = A_B/(2)^{1/2}$
54. (A) By $f = n[v/(2L)]$ & $v = [F_T/(m/L)]^{1/2} \Rightarrow f = (n/2)[F_T/(mL)]^{1/2} = (1/2)\{[(50)]/[(0.25)(5.5)]\}^{1/2} = 3.02 \text{ Hz}$
55. (A) By $E = \int dE = \int (k dq/r^2)$ & $dq = \lambda dx$ & $r = (12-x) \Rightarrow E = k\lambda \int (12-x)^{-2} dx = k\lambda (12-x)^{-1}|_0^{10} = (8.99\text{E}+9)(6.5\text{E}-9)(1/2 - 1/12) = 24 \text{ N/C}$
56. (D) By $W_{\text{net}} = W_{\text{ext}} + W_E = \Delta K = 0 \Rightarrow W_{\text{ext}} = -W_E = -(\Delta U_E) = +\Delta U_E = kq_1q_2/r - 0 = (8.99\text{E}+9)(+5\text{E}-6)(+4\text{E}-6)/(6) = 3.0\text{E}-2 \text{ J}$
57. (B) For series combinations: $C_{\text{eq}} = [(20\mu\text{F})^{-1} + (25\mu\text{F})^{-1}]^{-1} = 11.11\mu\text{F}$ and since the charge is the same in series: $Q = Q_{20} = Q_{25} = C_{\text{eq}}V = (11.11\text{E}-6)(12) = 133 \mu\text{C}$
58. (D) By $\mathbf{F}_B = q\mathbf{v} \times \mathbf{B} = (-1.602\text{E}-19)[(3.0 \times 10^6\mathbf{i} + 2.0 \times 10^6\mathbf{j}) \times (0.65\mathbf{i} - 0.45\mathbf{j} + 0.35\mathbf{k})]$
 $= (-1.12\text{E}-13\mathbf{i} + 1.68\text{E}-13\mathbf{j} + 4.25\text{E}-13\mathbf{k}) \text{ N}$
59. (C) By Ampère’s Law: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} \Rightarrow B = \mu_0 NI/(2\pi R)$ & $N = 1200/[2\pi(0.025)] = 7639$
 $\Rightarrow B = [(4\pi\text{E}-7)(7639)(300)]/[2\pi(0.3)] = 1.53 \text{ T}$
60. (D) With $F = IA/c = P/c$ and double for reflection gives:
 $(1/2 P)/c + 2(1/2 P)/c = (3/2)(25000)/(3\text{E}+8) = 1.3\text{E}-4 \text{ N}$



SCIENCE

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GENERAL DIRECTIONS:

- DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
- Ninety minutes should be ample time to complete this contest, but since it is not a race, contestants may take up to two hours. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet.
- You may place as many notations as you desire anywhere on the test paper except on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- On the back of this page is printed a copy of the periodic table of the elements. You may wish to refer to this table in answering the questions, and if needed, you may use the atomic weights and atomic numbers from the table. Other scientific relationships are listed also.
- Silent hand-held calculators that do not need external wall plugs may be used. Graphing calculators that do not have built-in or stored functionality that provides additional scientific information are allowed. Small hand-held computers are not permitted. Calculators that accept memory cards or memory sticks are not permitted. Each contestant may bring one spare calculator. All memory must be cleared.

SCORING:

All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for an incorrect answer.

Periodic Table of the Elements

1A 1		2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.008																		2 He 4.003
3 Li 6.941	4 Be 9.012												5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12		13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41		31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41		49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59		81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (285)							

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Some Standard Properties of Water

property	symbol	value
density of water	ρ_{water}	1.000 g cm ⁻³
density of ice	ρ_{ice}	0.9167 g cm ⁻³
specific heats		
ice	C_{ice}	2.09 J g ⁻¹ K ⁻¹
water	C_{water}	4.184 J g ⁻¹ K ⁻¹
steam	C_{steam}	2.03 J g ⁻¹ K ⁻¹
heat of fusion	ΔH_{fus} or L_f	334 J g ⁻¹
heat of vaporization	ΔH_{vap} or L_v	2260 J g ⁻¹
index of refraction	n	1.33
autoionization	K_w	1.0×10^{-14}

Pressure		
1 atm	=	760 torr
	=	101325 Pa
	=	14.7 psi
1 bar	=	10 ⁵ Pa
	=	100 kPa

Energy		
1 cal	=	4.184 J
1 L atm	=	101.325 J
1 Cal	=	1 kcal
1 hp	=	746 W
1 eV	=	1.602×10^{-19} J

Various Physical Constants

property	symbol	value
universal gas constant	R	8.314 J mol ⁻¹ K ⁻¹ 62.36 L torr mol ⁻¹ K ⁻¹ 0.08206 L atm mol ⁻¹ K ⁻¹ 1.987 cal mol ⁻¹ K ⁻¹
Planck's constant	h	6.626×10^{-34} J s 4.136×10^{-15} eV s
Planck's reduced constant	$h/2\pi$	1.054×10^{-34} J s 6.582×10^{-16} eV s
Boltzmann constant	k_B	1.38×10^{-23} J K ⁻¹
Stefan-Boltzmann	σ	5.67×10^{-8} W m ⁻² K ⁻⁴
speed of light	c	3.00×10^8 m s ⁻¹
speed of sound (at 20°C)	v_{air}	343 m s ⁻¹
acceleration of gravity	g	9.80 m s ⁻²
gravitational constant	G	6.67×10^{-11} N m ² kg ⁻²
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
elementary charge	e	1.602×10^{-19} C
Faraday	F	96485 C mol ⁻¹
Coulomb's law constant	k	8.988×10^9 N m ² C ⁻²
Rydberg constant	R	2.178×10^{-18} J

Some Other Conversion Factors

1 in	=	2.54 cm
1 lb	=	453.6 g
1 mi	=	5280 ft = 1.609 km
1 gal	=	4 quarts = 231 in ³ = 3.785 L

property	symbol	value
electron rest mass	m_e	9.11×10^{-31} kg 0.000549 u 0.511 MeV c ⁻²
proton mass	m_p	1.6726×10^{-27} kg 1.00728 u 938.3 MeV c ⁻²
neutron mass	m_n	1.6749×10^{-27} kg 1.008665 u 939.6 MeV c ⁻²
atomic mass unit	u	1.6605×10^{-27} kg 931.5 MeV c ⁻²
earth mass		5.972×10^{24} kg
earth radius		6.371×10^6 m
moon mass		7.348×10^{22} kg
sun mass		1.989×10^{30} kg
distance earth-moon		3.844×10^8 m
distance earth-sun		1.496×10^{11} m
permittivity of free space	ϵ_0	8.85×10^{-12} F m ⁻¹
permeability of free space	μ_0	$4\pi \times 10^{-7}$ T m A ⁻¹

Some Average Bond Energies (kJ/mol)

C-H	413	C-C	346	C-Cl	339	C-N	305
O-H	463	C=C	602	C-Br	285	N=N	418
N-H	391	C≡C	835	O=O	498	H-H	436
C-O	358	C=O	799	C≡O	1072	Br-Br	193
H-Cl	432	S-H	347	N≡N	945	Cl-Cl	242
H-Br	366	H-I	299	C≡N	887	I-I	151

Biology Questions (1 – 20)

1. Post-translational modifications of a polypeptide may include which of the following?
 - A) condensation reactions
 - B) hydrolytic reactions
 - C) hydrostatic reactions
 - D) A and B
 - E) A, B and C
2. The process of photosynthesis requires carbon, hydrogen and oxygen and sunlight energy. How do plants obtain the oxygen for photosynthesis?
 - A) from H₂O in the soil
 - B) from O₂ in the air
 - C) from CO₂ in the air
 - D) from glycolysis in the cytoplasm
 - E) from sunlight energy
3. Which of the following are considered a post-zygotic isolating mechanism in angiosperms?
 - A) pollen grain unable to germinate on the stigma
 - B) pollen tube unable to grow through the style
 - C) sperm nuclei unable to fertilize the egg cell
 - D) All the above
 - E) None of the above
4. Which of the following can be found in a bacterial chromosome?
 - A) histone proteins
 - B) telomeres
 - C) centromeres
 - D) DNA
 - E) All the above
5. Allopatric populations and sympatric populations of the same species must differ in their degree of:
 - A) gene flow
 - B) directional selection pressure
 - C) population sizes
 - D) mutation
 - E) genetic drift
6. Evolutionary change in which a parent species gives rise to many new species driven by availability of ecological niches is called
 - A) allopatric speciation.
 - B) sympatric speciation.
 - C) ecological diversification.
 - D) adaptive radiation.
 - E) punctuated adaptation.
7. Which is a main difference between cyclic and non-cyclic electron flow in photosynthesis?
 - A) Reduction of FAD
 - B) Reduction of NADP+
 - C) Reduction of ADP
 - D) Generation of ATP
 - E) Phosphorylation of ADP
8. The eukaryotic gene is divided into coding regions called exons and non-coding regions called introns. What is main difference between exons and introns?
 - A) Exons are transcribed and translated whereas introns are only transcribed.
 - B) Introns are transcribed and translated whereas exons are only transcribed.
 - C) Introns are never transcribed but exons are always transcribed.
 - D) Exons are located in within a gene whereas introns are located between genes.
 - E) Exons are RNA and introns are DNA.
9. The ecological diversity of insects can be attributed primarily to which of the following adaptations of the insects?
 - A) Complete and incomplete metamorphosis
 - B) Rigid exoskeleton made of chitin
 - C) Specialized mouthparts and hinged appendages
 - D) Extreme temperature tolerance
 - E) Ability to digest cellulose
10. *Taq* polymerase is an enzyme used in PCR that functions best at 75°C. From which group of organisms was the gene for *Taq* isolated?
 - A) E. coli bacteria
 - B) Thermophilic bacteria
 - C) Halophilic bacteria
 - D) Halophilic archaeans
 - E) Thermophilic protists

11. Which of the following is a true statement about the main difference between purine and pyrimidine nitrogenous bases?
 - A) purines contain uracil
 - B) pyrimidines are found in both DNA and RNA, purines are only found in DNA
 - C) purines have single ring structure whereas pyrimidines have a dual ring structure
 - D) purines can only form a chemical bond with other purines whereas pyrimidines can form a chemical bond with purines and pyrimidines
 - E) there are three types of pyrimidines and only two types of purines found in nucleic acids
12. Connective tissue holds together organs and tissues in an extracellular matrix. The matrix consists of a web of fibers embedded in a diverse medium. Which type of fibers can be found in connective tissue?
 - A) fibroblast, macrophage, and platelet
 - B) pith, soluble and extracellular
 - C) adipose, elastic and tendon
 - D) collagenous, reticular, and elastic
 - E) chondrocyte, erythrocyte and leukocyte
13. Frederick Griffith's 1928 experiments on transformation in Streptococcus pneumonia demonstrated which of the following?
 - A) DNA, not RNA is the genetic material of a chromosome.
 - B) DNA, not proteins, is the genetic material of a phage.
 - C) Changes in genotype and phenotype in a cell line can occur by the assimilation of external DNA.
 - D) The regularity in the ratio of nucleotide bases in DNA.
 - E) The double helix structure of DNA.
14. Which of the following organelles would be found in great concentration in cells found in the testes or ovaries?
 - A) Rough endoplasmic reticulum
 - B) Smooth endoplasmic reticulum
 - C) Ribosomes
 - D) Golgi apparatus
 - E) Vacuoles
15. The coupling of the potential energy of a proton gradient to the formation of a phosphoanhydride bond occurs in which of the following?
 - A) mitochondria
 - B) bacteria
 - C) nucleus
 - D) A and B
 - E) A, B and C
16. Non-protein components of enzymes can be classified as cofactors or coenzymes. The main difference between cofactors and coenzymes is
 - A) cofactors are inorganic ions and coenzymes are organic molecules.
 - B) coenzymes are not a component of vesicular proteins.
 - C) cofactors are found in prokaryotes and coenzymes are found in eukaryotes.
 - D) coenzymes help determine the tertiary structure and cofactors help determine the quaternary structure of proteins.
17. Pyruvate decarboxylase is an enzyme that does which of the following?
 - A) reduces acetaldehyde to alcohol in fermentation
 - B) removes a COOH group from a pyruvate molecule
 - C) attaches a carboxyl group to a pyruvate molecule
 - D) oxidizes a carboxyl group
 - E) reduces pyruvate
18. In a population under Hardy-Weinberg equilibrium, calculate the frequency of heterozygotes when the dominant allele at the locus is fixed.
 - A) $=2(1)(0)$
 - B) $=0.5^2$
 - C) Cannot be determined without knowing the frequency of the recessive allele.
 - D) $=1$
 - E) $=2(0.5)(0.5)$
19. A strictly vegan diet excludes meat, dairy, fish, eggs and poultry and all food that contain these animal products. Which of the following essential dietary item is lacking in a vegan diet?
 - A) cellulose
 - B) hydrolytic enzymes
 - C) vitamin B-12
 - D) vitamin C
 - E) protein

20. When building a phylogeny using molecular homologies, which of the following groupings of taxa is most desired?

A) monophyletic
B) paraphyletic
C) polyphyletic
D) analogous
E) heterologous

25. Under standard conditions, which one of the following reactions will have a value of ΔH that is equal to the enthalpy of formation for the compound formed?

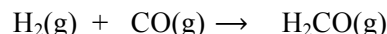
A) $4\text{Al(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{Al}_2\text{O}_3\text{(s)}$
B) $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO(g)}$
C) $\text{Pb}^{2+}\text{(aq)} + 2\text{Cl}^-\text{(aq)} \rightarrow \text{PbCl}_2\text{(s)}$
D) $\text{H}_2\text{O(l)} + \text{CO}_2\text{(g)} \rightarrow \text{H}_2\text{CO}_3\text{(aq)}$
E) $\frac{1}{2}\text{N}_2\text{(g)} + \frac{3}{2}\text{H}_2\text{(g)} \rightarrow \text{NH}_3\text{(g)}$

Chemistry Questions (21 – 40)

21.



26. Estimate the value for ΔH for the following gas phase reaction producing formaldehyde:

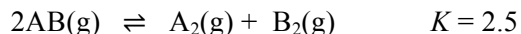


A) -390 kJ D) +324 kJ
B) -183 kJ E) -117 kJ
C) +296 kJ

22. In general, double bonds are _____ and _____ than single bonds.

A) longer ; stronger
B) longer ; weaker
C) shorter ; weaker
D) shorter ; stronger
E) none of the above

27. Consider the following reaction and its value for K .



Each of the three gases is currently at a pressure of 1.5 bar. Which of the following best describes what will happen next?

A) reaction will reverse to make more AB
B) reaction will go forward to make more A_2 and B_2
C) reaction does nothing because it is at equilibrium
D) reaction will go forward to make more AB
E) reaction will reverse to make more A_2 and B_2

23. The limits of the human eye can detect light with a minimum total energy of 3.8×10^{-17} J. If we assume that the light has a wavelength of 575 nm, how many photons does this correspond to?

A) 125 C) 110 E) 330
B) 85 D) 175

28. The vapor pressure of pure benzene (C_6H_6) at 25°C is 100 torr. A solution is made by dissolving 12.68 g of compound X (63.40 g/mol) into 100 g of benzene. What is the vapor pressure of this solution?

A) 88.7 torr D) 92.6 torr
B) 15.1 torr E) 67.9 torr
C) 86.5 torr

24. Which of the following compounds is capable of hydrogen-bonding?

A) $\text{CH}_3\text{CH}_2\text{CH}_3$ D) BeH_2
B) $\text{CH}_3\text{CH}_2\text{COOH}$ E) KF
C) HBr

29. A 0.084 M solution of a weak acid is 4.2% ionized. What is the pH of this solution?

A) 3.11 C) 1.08 E) 1.23
B) 1.45 D) 2.45

30. Which of the following combinations will make 100 mL of a working buffer?
- 65 mL of 0.1 M NH_4Cl + 35 mL of 0.1 M NaOH
 - 50 mL of 0.1 M HCl + 50 mL of 0.1 M NaOH
 - 75 mL of 0.1 M HCOOH + 25 mL of 0.1 M HCl
 - 50 mL of 0.1 M NH_3 + 50 mL of 0.1 M HCl
 - 40 mL of 0.1 M KOH + 60 mL of 0.1 M HNO_3

31. ZnCO_3 is fairly insoluble in water. Which of the following solutions could be added to water in order to help completely dissolve the ZnCO_3 ?

- a solution of sodium carbonate
- a 50/50 mix of ethanol and water
- a solution of potassium hydroxide
- a solution of hydrochloric acid
- a solution of silver nitrate

32. For the second order reaction $\text{X} \rightarrow 2\text{Y}$, it took 3 hours for 0.425 M X to drop to 0.108 M. What is the rate constant for this reaction?

- $6.39 \times 10^{-4} \text{ M}^{-1} \text{ s}^{-1}$
- $1.27 \times 10^{-4} \text{ M}^{-1} \text{ s}^{-1}$
- $1.34 \times 10^{-5} \text{ M}^{-1} \text{ s}^{-1}$
- $5.12 \times 10^{-4} \text{ M}^{-1} \text{ s}^{-1}$
- $2.94 \times 10^{-5} \text{ M}^{-1} \text{ s}^{-1}$

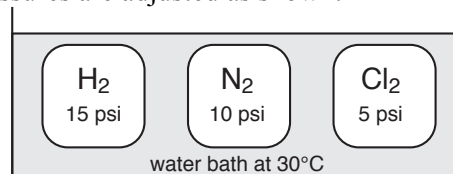
33. Which of the following elements is diamagnetic in its ground state electron configuration?

- F
- Na
- Si
- Pb
- Sr

34. Drops of 1.0 M KOH are added to a solution of a weak acid, HX, until the pH has been brought to exactly 5. The K_a value for HX is 3.2×10^{-5} . Which of the following solution species has the maximum concentration in this solution?

- HX
- OH^-
- X^-
- H_3O^+
- H_2X^+

35. Three identical containers are each filled with hydrogen gas, nitrogen gas, and chlorine gas while submerged in a water bath held at 30°C . The pressures are adjusted as shown.



Which container has the gas with the highest average kinetic energy per mole of gas?

- the H_2 container
- the N_2 container
- the Cl_2 container
- they all have the same kinetic energy

36. A solution of sodium sulfate can be electrolyzed to produce oxygen gas and hydrogen gas at the two electrodes. Which of the following is true for this electrolytic cell at the surface of the anode?

- H_2 gas is produced and the pH is raised
- O_2 gas is produced and the pH is lowered
- H_2 gas is produced and the pH is lowered
- O_2 gas is produced and the pH is raised

37. Under standard thermodynamic conditions, which of the following is the correct mathematical relationship comparing enthalpy (H) to internal energy (U) for *any* given system?

- $H = U$
- $H = 1/U$
- $H < U$
- $H < 0 < U$
- $H > U$

38. Consider the molecular structure for the chloroformate ion, ClCO_2^- . The two carbon-oxygen bonds have _____ bond strengths and _____ bond lengths.

- different ; the same
- different ; different
- the same ; the same
- the same ; different

39. All of the following chemical species have the same molecular geometry except one. Which one is different?
- A) CO_3^{2-} C) SO_3 E) CCl_2O
 B) PO_3^{3-} D) NO_3^-
40. According to atomic theory, what is the maximum number of electrons can have a principle quantum number of $n = 4$, and a magnetic quantum number of $m_\ell = +1$?
- A) 6 C) 1 E) 15
 B) 7 D) 3
43. According to Tyson, which of the following is not included in the IAU's criterion for a planet?
- A) is in orbit around a star, but may orbit a spot that is exterior to the object
 B) is in orbit around a star, but not in orbit around another planet
 C) it has cleared its orbit of wayward debris
 D) large enough for its own force of gravity to shape it into a sphere
 E) not so large that it would trigger fusion in its core
44. According to Tyson, which scientist is attributed with saying "the stated authority of a thousand is not worth the humble reasoning of a single individual"?
- A) Nicolaus Copernicus
 B) Galileo Galilei
 C) David Jewitt
 D) Bill Nye the Science Guy®
 E) Neil DeGrasse Tyson

Physics Questions (41 – 60)

On this contest $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors in the $+x$, $+y$ and $+z$ directions respectively, and unless otherwise indicated, you may neglect air resistance in every problem.

41. According to Tyson, in the Planet Zone of the Hall of the Universe at the Rose Center which of the following is NOT included as one of the five classes of objects that orbit our Sun?
- A) asteroid belt
 B) dwarf planets
 C) gas giant planets
 D) Kuiper Belt
 E) terrestrial planets
42. According to Tyson, who opined "The great thing about [the Pluto] debate is that it has gotten people everywhere thinking about planets and our place among them."?
- A) David Levy
 B) Bill Nye the Science Guy®
 C) Geoff Marcy
 D) Mark Sykes
 E) Neil DeGrasse Tyson
45. A rocket is launched straight up from rest at ground level with a constant acceleration of 23 m/s^2 for 35 s until the engine runs out of fuel. After 35 s, the rocket continues to rise upward moving now in free fall. What is the highest point above ground level reached by the rocket?
- A) $1.4 \times 10^4 \text{ m}$
 B) $1.9 \times 10^4 \text{ m}$
 C) $2.8 \times 10^4 \text{ m}$
 D) $3.2 \times 10^4 \text{ m}$
 E) $4.7 \times 10^4 \text{ m}$
46. Two figure skaters, Kate with a mass of 95 kg and Robert with a mass of 150 kg, are at rest in the middle of the ice rink. If Robert pushes Kate with an average force of 26 N for 2.0 s, then what is the speed of Robert after he finishes pushing Kate? You may neglect friction between the skaters and the ice.
- A) 0.00 m/s
 B) 0.21 m/s
 C) 0.27 m/s
 D) 0.35 m/s
 E) 0.55 m/s

47. Find the angle between vectors $\mathbf{A}$ and $\mathbf{B}$ given $\mathbf{A} = 3.0\mathbf{i} + 5.0\mathbf{j}$ and $\mathbf{B} = -4.0\mathbf{i} + 3.0\mathbf{j}$.
- 21°
 - 53°
 - 59°
 - 84°
 - 90°
48. A solid, uniform sphere with a mass of 2.7 kg and a diameter of 20 cm rotates about an axis tangent to the surface of the sphere at a constant 50 rev/min. What is the rotational kinetic energy of the sphere?
- 0.10 J
 - 0.15 J
 - 0.52 J
 - 0.70 J
 - 1.4 J
49. A projectile of mass m is launched with an initial velocity v_0 and at an angle θ above the horizontal. In a coordinate system with the origin at the launch point and oriented such that the plane of the motion is described by $\mathbf{i}$ and $\mathbf{j}$ with $\mathbf{i}$ pointing in the horizontal direction of motion and with $\mathbf{j}$ pointing vertically upward. What is the time rate of change of the angular momentum vector about the launch point as a function of the time from launch, t ?
- $[-mgt \sin\theta]\mathbf{i}$
 - $[-mgtv_0 \sin\theta]\mathbf{k}$
 - $[-mgtv_0 \cos\theta]\mathbf{k}$
 - $[mgt \sin\theta]\mathbf{i}$
 - $[mgv_0 \cos\theta]\mathbf{i}$
50. A steel wire is under tension of 47.0 N. The wire is 0.150 mm in diameter and has a length under tension of 40.0 cm. Given that the elastic or Young's modulus of steel is 200 GPa, then what is the length of the wire without any tension?
- 39.0 cm
 - 39.5 cm
 - 40.0 cm
 - 41.0 cm
 - 52.5 cm
51. Water at 20°C undergoes laminar flow in a horizontal pipe that has a radius of 2.0 cm and length of 20 m with a pressure drop 22 kPa across the ends of the pipe. Given that the viscosity of water at 20°C is 1.00 mPa·s, then what is the speed of water flowing through the pipe? You may treat the water as incompressible and Newtonian.
- $5.5 \times 10^1 \text{ m/s}$
 - $1.4 \times 10^2 \text{ m/s}$
 - $4.4 \times 10^2 \text{ m/s}$
 - $3.5 \times 10^3 \text{ m/s}$
 - $1.1 \times 10^4 \text{ m/s}$
52. Two tuning forks are struck and the sounds from each fork reach your ears simultaneously with equal intensities. One fork has a frequency of 243 Hz and the second fork has a frequency of 245 Hz. What is the frequency (or pitch) of the tone that is heard?
- 2 Hz
 - 243 Hz
 - 244 Hz
 - 245 Hz
 - 488 Hz
53. For a molecular gas which of the following quantities is determined solely by the average translational kinetic energy of the molecules of the gas?
- enthalpy
 - entropy
 - internal energy
 - pressure
 - temperature
54. The volume charge density, ρ , of a solid, non-conducting sphere of radius R varies with radial distance r as given by $\rho(r) = A/r$, where A is constant. What is the total charge of the sphere?
- $A \ln(R)$
 - $A/(2R^2)$
 - $4\pi A$
 - πAR^2
 - $2\pi AR^2$

55. Two point charges with a positive charge, $+q$, are located at $(+a, 0)$ and $(-a, 0)$. What are the values of the net electric field and the net electric potential at the origin? You may take the electric potential to be zero infinitely far from the charges.
- $\Sigma \mathbf{E} = 0$ and $\Sigma V = 0$
 - $\Sigma \mathbf{E} = 0$ and $\Sigma V = 2kq/a$
 - $\Sigma \mathbf{E} = (2kq/a^2)\mathbf{i}$ and $\Sigma V = 0$
 - $\Sigma \mathbf{E} = (2kq/a^2)\mathbf{i}$ and $\Sigma V = 2kq/a$
 - none of the above
56. A parallel-plate capacitor has a dielectric material inserted between the conductive plates such that it completely fills the space between the plates. The area of each plate is 250 cm^2 and the plates are 0.300 mm apart. If a battery charges the capacitor to a potential difference of 120 V and places equal amounts of opposite free charge of magnitude 150 nC on each plate, then what is the magnitude of the bound charge induced on each dielectric surface?
- 0.00 nC
 - 61.5 nC
 - 75.0 nC
 - 88.5 nC
 - 150 nC
57. An alpha particle with charge $+2e$ travels in a circular path with a radius of 1.25 m in a uniform magnetic field with a magnitude of 0.25 T . Given that the mass of an alpha particle is $6.65 \times 10^{-27} \text{ kg}$, what is the period of the circular motion?
- $3.3 \times 10^{-9} \text{ s}$
 - $4.2 \times 10^{-8} \text{ s}$
 - $1.0 \times 10^{-8} \text{ s}$
 - $5.2 \times 10^{-7} \text{ s}$
 - $1.7 \times 10^{-7} \text{ s}$
58. Two long, straight wires lie in a horizontal plane and carry equal currents in opposite directions. If the wire that has the current that flows east is located north of the wire that has the current that flows west, then at a point halfway between the wires in the horizontal plane what is the direction of the net magnetic field from the wires? You may neglect the magnetic field of the Earth.
- vertically down
 - vertically up
 - north
 - south
 - zero
59. A long, thin wire carries a current of 10 A . A long, hollow, cylindrical conductor is oriented so that it is coaxial to the wire and has an inner radius of 1.5 mm and an outer radius of 3.0 mm . The cylindrical conductor carries a current of 12 A that is uniformly distributed over its cross-sectional area and flows in the opposite direction as the current in the wire. What is the magnitude of the magnetic field at a distance of 2.0 mm from the common central axis of the wire and the cylinder?
- $2.0 \times 10^{-4} \text{ T}$
 - $6.9 \times 10^{-4} \text{ T}$
 - $1.0 \times 10^{-3} \text{ T}$
 - $2.0 \times 10^{-3} \text{ T}$
 - $2.2 \times 10^{-3} \text{ T}$
60. The objective lens of a compound optical (or light) microscope has a focal length of 22.0 mm . It forms an image of a specimen at a distance of 18.0 cm beyond the focal point located on the same side as the image. How far from the objective lens is the specimen? You may neglect the thickness of the lens and use the thin lens approximation.
- 2.20 cm
 - 2.47 cm
 - 18.0 cm
 - 20.2 cm
 - 44.4 cm

UIL HIGH SCHOOL SCIENCE CONTEST

Contestant Answer Sheet

Contestant # _____

9 10 11 12
CONTESTANT GRADE LEVEL

A 2A 3A 4A 5A
CONFERENCE

PLEASE WRITE ANSWERS WITH CAPITAL LETTERS

- | | | |
|-----------|-----------|-----------|
| 1. _____ | 21. _____ | 41. _____ |
| 2. _____ | 22. _____ | 42. _____ |
| 3. _____ | 23. _____ | 43. _____ |
| 4. _____ | 24. _____ | 44. _____ |
| 5. _____ | 25. _____ | 45. _____ |
| 6. _____ | 26. _____ | 46. _____ |
| 7. _____ | 27. _____ | 47. _____ |
| 8. _____ | 28. _____ | 48. _____ |
| 9. _____ | 29. _____ | 49. _____ |
| 10. _____ | 30. _____ | 50. _____ |
| 11. _____ | 31. _____ | 51. _____ |
| 12. _____ | 32. _____ | 52. _____ |
| 13. _____ | 33. _____ | 53. _____ |
| 14. _____ | 34. _____ | 54. _____ |
| 15. _____ | 35. _____ | 55. _____ |
| 16. _____ | 36. _____ | 56. _____ |
| 17. _____ | 37. _____ | 57. _____ |
| 18. _____ | 38. _____ | 58. _____ |
| 19. _____ | 39. _____ | 59. _____ |
| 20. _____ | 40. _____ | 60. _____ |

BIOLOGY SCORE

CHEMISTRY SCORE

PHYSICS SCORE

OVERALL SCORE:

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY**

REGION • 2015

- | | | |
|-------|---|-------|
| 1. D | 21.  | 41. B |
| 2. C | 22. D | 42. B |
| 3. E | 23. C | 43. A |
| 4. D | 24. B | 44. B |
| 5. A | 25. E | 45. E |
| 6. D | 26. E | 46. D |
| 7. B | 27. B | 47. D |
| 8. A | 28. C | 48. C |
| 9. A | 29. D | 49. C |
| 10. B | 30. A | 50. B |
| 11. E | 31. D | 51. A |
| 12. D | 32. A | 52. C |
| 13. C | 33. E | 53. E |
| 14. B | 34. C | 54. E |
| 15. D | 35. D | 55. B |
| 16. A | 36. B | 56. B |
| 17. B | 37. E | 57. D |
| 18. A | 38. C | 58. A |
| 19. C | 39. B | 59. B |
| 20. A | 40. A | 60. B |

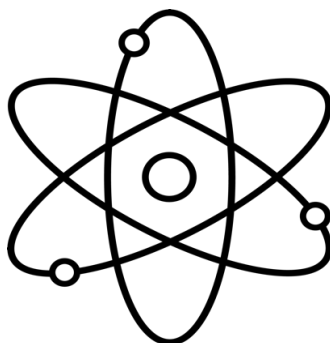
PHYSICS KEY for Science Contest • Region • 2015

41. (B) “Five classes of objects orbit our Sun. ... Beyond the outer planets is the **Kuiper Belt** of comets, a disk of small icy worlds including Pluto.”
42. (B) The quote is from Bill Nye the Science Guy® which is found in Chapter 5.
43. (A) This is not one of the IAU’s criteria for a planet as discussed in Chapter 6.
44. (B) “As is often cited (and attributed to Galileo), the stated authority of a thousand is not worth the humble reasoning of a single individual.”
45. (E) By kinematics for the first 35 s: $(y - y_0)_1 = v_0 t + \frac{1}{2} a t^2 = 0 + \frac{1}{2} (+23)(35)^2 = 1.41\text{E}+4 \text{ m}$ & $v = v_0 + at = 0 + (+23)(35) = 805 \text{ m/s}$. Then for the free fall part of the motion: $(y - y_0)_2 = [v^2 - v_0^2]/[2a] = [0^2 - 805^2]/[2(-9.8)] = 3.31\text{E}+4 \text{ m} \Rightarrow h = (y - y_0)_1 + (y - y_0)_2 = 1.41\text{E}+4 + 3.31\text{E}+4 = 4.7\text{E}+4 \text{ m}$
46. (D) By $\Delta p = (\Sigma F)_{\text{avg}} \Delta t \Rightarrow v = (26)(2)/(150) = 0.346 \text{ m/s}$
47. (D) By the formula for the dot product: $\theta = \cos^{-1}([A \cdot B]/[AB]) = \cos^{-1}([(3)(-4) + (5)(3)]/[\{(3)^2 + (5)^2 \}^{1/2} \{(-4)^2 + (3)^2 \}^{1/2}]) = 84^\circ$
48. (C) By the parallel axis theorem: $I = I_{\text{com}} + Md^2 = (2/5)MR^2 + MR^2 = (7/5)MR^2$, and $KE_{\text{rot}} = \frac{1}{2}I\omega^2 = \frac{1}{2}[(7/5)(2.7)(0.1)^2][(50)(2\pi/60)]^2 = 0.52 \text{ J}$
49. (C) With $\mathbf{r}(t) = [(v_0 \cos \theta)t] \mathbf{i} + [(v_0 \sin \theta)t - \frac{1}{2}gt^2] \mathbf{j} \Rightarrow \mathbf{p}(t) = m \{ [(v_0 \cos \theta)] \mathbf{i} + [(v_0 \sin \theta) - gt] \mathbf{j} \} \Rightarrow \mathbf{L}(t) = \mathbf{r}(t) \times \mathbf{p}(t) = [-\frac{1}{2}mgt^2 v_0 \cos \theta] \mathbf{k} \Rightarrow (d/dt)\mathbf{L}(t) = [-mgt v_0 \cos \theta] \mathbf{k}$, which is also equal to the net torque that is produced solely by the gravitational force.
50. (B) By $F/A = Y\Delta L/L_0 \Rightarrow L_0 = L \{ F/(YA) + 1 \}^{-1} = (0.40) \{ (47)/[\pi(0.075\text{E}-3)^2(200\text{E}+9)] + 1 \}^{-1} = 0.395 \text{ m}$
51. (A) By Poiseuille’s law: $\Delta P = [(8\eta L)/(\pi r^4)]Q$, where the volume flowrate $Q = Av = (\pi r^2)v \Rightarrow v = [(r^2 \Delta P)/(8\eta L)] = [(0.02)^2(22\text{E}+3)]/[8(1\text{E}-3)(20)] = 55 \text{ m/s}$
52. (C) The frequency that is heard is the average of the two individual frequencies $f = (243+245)/2 = 244 \text{ Hz}$. Note that the intensity of this frequency is modulated by the beat frequency.
53. (E) By definition the temperature of a molecular gas is the average translational kinetic energy of the molecules of the gas.
54. (E) $Q = \int \rho(r) dV$, where $dV = (4\pi r^2) dr \Rightarrow Q = \int_0^R (A/r) (4\pi r^2) dr = 2\pi AR^2$
55. (B) At the midpoint between two identical charges the net electric field is zero and the electric potential from each charge add to give $\Sigma V = 2kq/a$
56. (B) Without the dielectric $C_0 = \epsilon_0 A/d = (8.85\text{E}-12)(250\text{E}-4)/(0.3\text{E}-3) = 8.85\text{E}-8 \text{ F}$ and thus $Q_0 = C_0 V = (8.85\text{E}-8)(120) = 88.5 \text{ nC} \Rightarrow Q_{\text{bound}} = Q - Q_0 = 150 - 88.5 = 61.5 \text{ nC}$
57. (D) By N2L in the centripetal direction: $\Sigma F_c = F_B = |q|vB \sin 90^\circ = mv^2/r$ and with $v = 2\pi r/T$
 $\Rightarrow T = [2\pi m]/[|q|B] = [2\pi(6.65\text{E}-27)]/[2(1.602\text{E}-19)(0.25)] = 5.2\text{E}-7 \text{ s}$
58. (A) By the right hand circulation rule, the direction of both magnetic fields from the wires and therefore the net magnetic field is vertically down.
59. (B) By Ampere’s law: $\int_{\text{loop}} \mathbf{B} \cdot d\mathbf{s} = \mu_0 i_{\text{enclosed}} \Rightarrow B(2\pi r) = \mu_0 \{ i_{\text{wire}} - i_{\text{cylinder}} [\pi(r^2 - r_{\text{inner}}^2)] / [\pi(r_{\text{outer}}^2 - r_{\text{inner}}^2)] \}$
 $\Rightarrow B = (4\pi\text{E}-7)/[2\pi(0.002)] \{ 10 - 12[(0.002^2 - 0.0015^2)/(0.003^2 - 0.0015^2)] \} = 6.9\text{E}-4 \text{ T}$
60. (B) The objective lens of a compound microscope is a converging lens that will form a real image that is then viewed by the eyepiece. Thus $f = +2.2 \text{ cm}$ and $d_i = 18 + 2.2 = +20.2 \text{ cm}$, which by $1/f = 1/d_o + 1/d_i \Rightarrow d_o = 2.47 \text{ cm}$



SCIENCE

State • 2015



GENERAL DIRECTIONS:

- DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
- Ninety minutes should be ample time to complete this contest, but since it is not a race, contestants may take up to two hours. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet.
- You may place as many notations as you desire anywhere on the test paper except on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- On the back of this page is printed a copy of the periodic table of the elements. You may wish to refer to this table in answering the questions, and if needed, you may use the atomic weights and atomic numbers from the table. Other scientific relationships are listed also.
- Silent hand-held calculators that do not need external wall plugs may be used. Graphing calculators that do not have built-in or stored functionality that provides additional scientific information are allowed. Small hand-held computers are not permitted. Calculators that accept memory cards or memory sticks are not permitted. Each contestant may bring one spare calculator. All memory must be cleared.

SCORING:

All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for an incorrect answer.

Periodic Table of the Elements

1A 1		2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.008																		2 He 4.003
3 Li 6.941	4 Be 9.012												5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12		13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41		31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41		49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59		81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (285)							

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Some Standard Properties of Water

property	symbol	value
density of water	ρ_{water}	1.000 g cm ⁻³
density of ice	ρ_{ice}	0.9167 g cm ⁻³
specific heats		
ice	C_{ice}	2.09 J g ⁻¹ K ⁻¹
water	C_{water}	4.184 J g ⁻¹ K ⁻¹
steam	C_{steam}	2.03 J g ⁻¹ K ⁻¹
heat of fusion	ΔH_{fus} or L_f	334 J g ⁻¹
heat of vaporization	ΔH_{vap} or L_v	2260 J g ⁻¹
index of refraction	n	1.33
autoionization	K_w	1.0×10^{-14}

Pressure		
1 atm	=	760 torr
	=	101325 Pa
	=	14.7 psi
1 bar	=	10 ⁵ Pa
	=	100 kPa

Energy		
1 cal	=	4.184 J
1 L atm	=	101.325 J
1 Cal	=	1 kcal
1 hp	=	746 W
1 eV	=	1.602×10^{-19} J

Various Physical Constants

property	symbol	value
universal gas constant	R	8.314 J mol ⁻¹ K ⁻¹ 62.36 L torr mol ⁻¹ K ⁻¹ 0.08206 L atm mol ⁻¹ K ⁻¹ 1.987 cal mol ⁻¹ K ⁻¹
Planck's constant	h	6.626×10^{-34} J s 4.136×10^{-15} eV s
Planck's reduced constant	$h/2\pi$	1.054×10^{-34} J s 6.582×10^{-16} eV s
Boltzmann constant	k_B	1.38×10^{-23} J K ⁻¹
Stefan-Boltzmann	σ	5.67×10^{-8} W m ⁻² K ⁻⁴
speed of light	c	3.00×10^8 m s ⁻¹
speed of sound (at 20°C)	v_{air}	343 m s ⁻¹
acceleration of gravity	g	9.80 m s ⁻²
gravitational constant	G	6.67×10^{-11} N m ² kg ⁻²
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
elementary charge	e	1.602×10^{-19} C
Faraday	F	96485 C mol ⁻¹
Coulomb's law constant	k	8.988×10^9 N m ² C ⁻²
Rydberg constant	R	2.178×10^{-18} J

Some Other Conversion Factors

1 in	=	2.54 cm
1 lb	=	453.6 g
1 mi	=	5280 ft = 1.609 km
1 gal	=	4 quarts = 231 in ³ = 3.785 L

property	symbol	value
electron rest mass	m_e	9.11×10^{-31} kg 0.000549 u 0.511 MeV c ⁻²
proton mass	m_p	1.6726×10^{-27} kg 1.00728 u 938.3 MeV c ⁻²
neutron mass	m_n	1.6749×10^{-27} kg 1.008665 u 939.6 MeV c ⁻²
atomic mass unit	u	1.6605×10^{-27} kg 931.5 MeV c ⁻²
earth mass		5.972×10^{24} kg
earth radius		6.371×10^6 m
moon mass		7.348×10^{22} kg
sun mass		1.989×10^{30} kg
distance earth-moon		3.844×10^8 m
distance earth-sun		1.496×10^{11} m
permittivity of free space	ϵ_0	8.85×10^{-12} F m ⁻¹
permeability of free space	μ_0	$4\pi \times 10^{-7}$ T m A ⁻¹

Some Average Bond Energies (kJ/mol)

C-H	413	C-C	346	C-Cl	339	C-N	305
O-H	463	C=C	602	C-Br	285	N=N	418
N-H	391	C≡C	835	O=O	498	H-H	436
C-O	358	C=O	799	C≡O	1072	Br-Br	193
H-Cl	432	S-H	347	N≡N	945	Cl-Cl	242
H-Br	366	H-I	299	C≡N	887	I-I	151

Biology Questions (1 – 20)

1. Herbaceous eudicots differ from woody eudicots in that
 - A) In herbaceous eudicots, the apical meristem no longer produces auxin.
 - B) In herbaceous eudicots, the lateral meristems are significantly less active.
 - C) In herbaceous eudicots, the cork cambium replaces the epidermis with periderm.
 - D) In herbaceous eudicots, the production of initials in meristems is by meiosis.
2. Innate immunity in animals is a defense system that is active immediately upon exposure to a pathogen. Adaptive immunity is found only in vertebrates and consists of which of the following items?
 - A) Skin, mucous membranes and secretions
 - B) Phagocytic cells
 - C) Inflammatory response
 - D) Cytotoxic cells
 - E) All the above are components of the adaptive immunity response.
3. The directional growth in plants in response to touch is called thigmotropism. The rapid leaf movement in the sensitive plant, *Mimosa pudica*, occurs as a result of the loss of turgor in the cells of the pulvinus, a specialized motor organ located at the joint of the leaf. Which of the following explains the rapid loss of turgor in pulvinus cells after stimulation?
 - A) Loss of K^+ followed by osmosis out of the cell.
 - B) Opening of H^+ pumps followed by osmosis into the cell.
 - C) Rapid elongation of cells due to auxin synthesis on one side of the pulvinus.
 - D) The rapid redistribution and active transport of statoliths across cells located in the pulvinus.
4. Small non-polar molecules like CO_2 and O_2 can diffuse across the plasma membrane. Large, polar, or charged molecules require a transport protein to move into and out of a cell. The diffusion of gasses limits cell size because:
 - A) the laws of physics dictate that diffusion is a slow process
 - B) the diffusion gradients require energy inputs provided by cellular respiration
 - C) the diffusion rate is proportional to the square of the distance of diffusion
 - D) the diffusion rate is inversely proportional the concentration gradient
 - E) gasses first must be dissolved in order to undergo diffusion
5. The cardiovascular system in vertebrates is composed of three main types of blood vessels: arteries, capillaries and veins. Choose the correct sequence of blood vessels as they carry blood away from and back to the heart.
 - A) Veins, venules, capillaries, arteries
 - B) Arterioles, arteries, capillaries, capillary beds, veins, venules
 - C) Arteries, capillaries, capillary beds, arterioles, venules, veins
 - D) Arteries, arterioles, capillary beds, capillaries, venules, veins
 - E) Arteries, arterioles, capillaries, capillary beds, venules, veins
6. After cellular infection, some viruses enter an inactive state called latency. Which of the following is NOT true regarding a dormant virus?
 - A) The viral genome continues to be replicated in infected cells.
 - B) Cells infected with the dormant virus cease to make most viral proteins.
 - C) Viral latency can be broken by environmental conditions.
 - D) The dormant virus continues to trigger the adaptive immune response.

7. Complete non-disjunction during meiosis in plants can lead to polyploidy, the doubling (or more) of chromosome sets. The main difference between allopolyploidy and autopolyploidy in plants is that in autopolyploidy the
 - A) chromosome sets are found in multiples of 3
 - B) chromosome sets arise from 2 different species
 - C) chromosome sets arise from 1 species
 - D) chromosome sets are the result of self-pollination
 - E) chromosome sets are the result of cross-pollination
8. X-inactivation in female mammals insures that females and males have the same effective dose of proteins coded by the X chromosome. Inactivation of the X chromosome involves which of the following processes?
 - A) Silencing of the promoters
 - B) Inhibition of the transcription factors
 - C) Acetylation of nucleosomes
 - D) Activation of the XIST gene
 - E) Degradation of the histones
9. The efficiency of signal transduction in cell signaling is increased by the presence of _____ which are large relay proteins to which several other relay proteins are simultaneously attached.
 - A) Scaffolding proteins
 - B) G-protein linked transducers
 - C) DAG proteins
 - D) Adenylyl cyclases
 - E) cAMPases
10. The scientist working with fruit flies who provided the most convincing evidence that chromosomes were responsible for Mendel's 'heritable factors' was
 - A) James Watson
 - B) Francis Crick
 - C) Frederick Griffith
 - D) Martha Chase
 - E) Thomas Hunt Morgan
11. An evolutionary change in the rate or timing of developmental events is known as
 - A) Evodevolution
 - B) Homoplasy
 - C) Heterochrony
 - D) Symplasy
 - E) Heteromorphism
12. A plate of ectodermal cells that roll into a hollow tube located dorsally to the notochord in all early embryonic forms in chordates
 - A) provides evidence for their shared ancestry
 - B) is present at all stages of life
 - C) is found only in vertebrates
 - D) develops into the bony vertebral column
 - E) is a result of convergent evolution
13. In plants, root hairs and trichomes are both derived from which plant tissue system?
 - A) Dermal
 - B) Ground
 - C) Vascular
 - D) Xylem
 - E) Phloem
14. Hydroponic cultures of plants are grown in mineral solutions without soil. The aerated solution that surrounds the roots must contain abundant dissolved oxygen to prevent which of the following?
 - A) Chlorosis of the leaves
 - B) Flood tolerance of the roots
 - C) Slowing of photosynthesis due to lack of oxygen
 - D) Alcoholic fermentation in the roots
 - E) Inactivation of proton pumps

15. Homeostasis in animals is a dynamic equilibrium that moderates changes in the internal environment. Which of the following plays the major role in homeostasis in animals?
- Amplified feedback
 - Dampened feedback
 - Negative feedback
 - Positive feedback
 - Thalamal feedback
16. There are four main feeding mechanisms found in animals: suspensions feeders, substrate feeders, fluid feeders and bulk feeders. Which of the following play a role in bulk feeding?
- Tentacles
 - Jaws
 - Baleen
 - A and B
 - A, B and C
17. As population density increases, density-dependent mechanisms stop or slow population growth. Which of the following illustrates a density- *independent* regulation of population growth?
- Intrinsic physiological factors that delay reproduction.
 - Predation that increases the death rate.
 - Production of toxic waste products that increase the death rate.
 - Competition for resources that decreases the birth rate.
 - Effects of natural disaster that decreases population size.
18. Measurements of biodiversity include which of the following?
- Genetic diversity within a population
 - Species diversity within a community
 - Ecosystem diversity with the biosphere
 - B and C
 - A, B and C
19. Which of the following accurately portrays the relationship between saturation of fatty acids and plasma membrane fluidity?
- As saturation increases, fluidity increases.
 - As saturation increases, fluidity decreases.
 - There is no discernable effect of saturation on membrane fluidity.
20. The ATP synthase is a molecular ‘mill’ that phosphorylates ADP to ATP using the power of a proton gradient in both the chloroplast and the mitochondrion. The ATP synthase has four main parts:
- the stator, rotor, rod and mill
 - the stator, rotor, mill and knob
 - the stator, mill, knob and rod
 - the stator, rod, mill and knob
 - the stator, rod, rotor and knob
- Chemistry Questions (21 – 40)**
21. Permanganate will oxidize chloride in acidic conditions to produce chlorine gas and manganese(II). The unbalanced reaction is
- $$\text{MnO}_4^-(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq}) + \text{Cl}_2(\text{g})$$
- 80 mL of 0.0500 M KMnO_4 is poured into 320 mL of 0.0750 M KCl . The resulting chlorine gas is collected at 722 torr and 28.0°C. What is the volume of the collected chlorine gas?
- | | |
|-----------|-----------|
| A) 325 mL | D) 312 mL |
| B) 195 mL | E) 260 mL |
| C) 173 mL | F) 624 mL |
22. Step 1. Balance the reaction:
- $$\text{CS}_2 + \text{O}_2 \rightarrow \text{CO}_2 + \text{SO}_2$$
- Step 2. What is the percent yield of this reaction when 62.3 mg of CS_2 reacts with excess oxygen to give 86.2 mg of SO_2 ?
- | | |
|-----------|-----------|
| A) 94.8 % | D) 79.1 % |
| B) 82.2 % | E) 91.4 % |
| C) 87.5 % | F) 98.3 % |

23. Use a table of bond energies to calculate the approximate heat of reaction between methanol and carbon monoxide to form acetic acid.

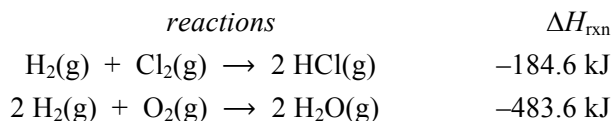


- A) -390 kJ D) -73 kJ
B) -183 kJ E) -117 kJ
C) +296 kJ

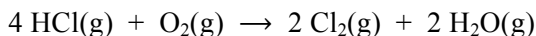
24. Benzene has a normal boiling point of 80°C and a heat of vaporization of 34 kJ/mol . If the applied pressure is changed to 100 torr, at what temperature will benzene begin to boil?

- A) 27°C D) 43°C
B) 32°C E) 51°C
C) 18°C F) 65°C

25. Given the following two reactions and the associated enthalpies of reaction:



Determine the value of ΔH_{rxn} for the following reaction:

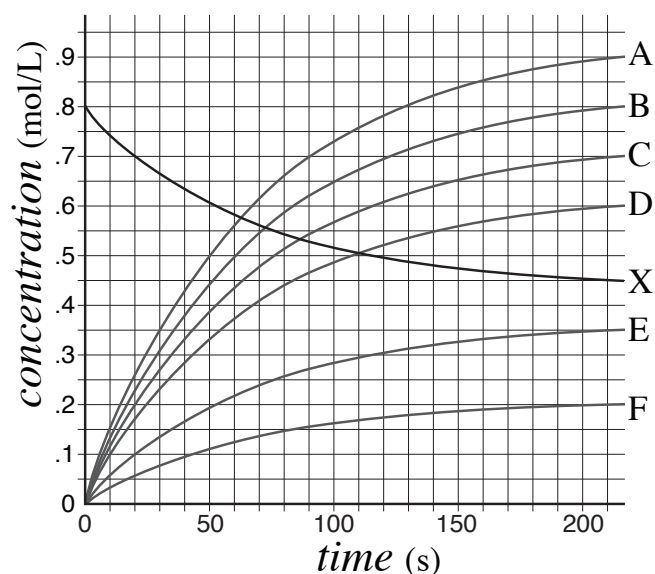


- A) -299.0 kJ D) +299.0 kJ
B) -598.0 kJ E) +598.0 kJ
C) -114.4 kJ F) +114.4 kJ

26. A weak acid, HX, is dissolved into water at a concentration of 0.30 m . The freezing point of this solution was precisely measured to be -0.614°C . The freezing point depression constant for water is $1.86^\circ\text{C}/\text{m}$. What is the percent ionization of this weak acid?

- A) 5% D) 10%
B) 33% E) 50%
C) 42% F) 25%

27. Below are several plots of concentration vs time. The plot labeled X is for the concentration of X in the reaction: $\text{X} \rightarrow 2\text{Y}$. Which one of the plots A-F is the correct plot for the concentration of Y during the same reaction?



28. Tris(hydroxymethyl)aminomethane is commonly known as “tris” and has a K_b equal to 1.2×10^{-6} . The formula for tris is $(\text{HOCH}_2)_3\text{CNH}_2$. 50 mL of 0.15 M hydrobromic acid is added to 200 mL of 0.17 M tris. What is the pH of the resulting 250 mL of solution?

- A) 7.53 C) 8.74 E) 7.42
B) 8.63 D) 8.08 F) 10.65

29. A concentration cell is set up for a generic metal and its cation:



What is the potential for this cell as written?

- A) -115 mV C) +230 mV E) +245 mV
B) -245 mV D) +115 mV F) -230 mV

30. The non-ionized form of an indicator is yellow while the ionized form is blue. The K_a for this indicator is 3.2×10^{-5} . One drop of this indicator is added to a 0.051 M solution of hypobromous acid (HOBr) which has a K_a of 2.3×10^{-9} . Which of the following best describes the color of this solution?

- A) blue D) blue-green
B) yellow-green E) yellow-orange
C) green F) yellow

31. What is the wavelength of light emitted when an electron in a hydrogen atom transitions from $n = 6$ to $n = 2$.

A) 128 nm C) 639 nm E) 411 nm
B) 274 nm D) 1250 nm F) 927 nm

32. According to quantum mechanics and atomic theory, which one of the following orbitals does not theoretically exist?

A) $7s$ C) $8p$ E) $6h$
B) $6d$ D) $5g$ F) $3f$

33. The following kinetic data was collected for the reaction: $A + 2B + C \rightarrow D$

exp	[A]	[B]	[C]	rate of formation of D (M/s)
1	0.15	0.10	0.20	1.25×10^{-3}
2	0.15	0.20	0.20	1.25×10^{-3}
3	0.15	0.30	0.40	2.50×10^{-3}
4	0.24	0.40	0.25	4.00×10^{-3}

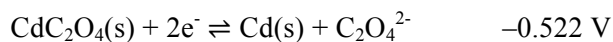
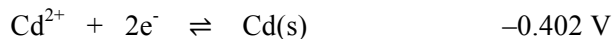
Which of the following equations is the rate law for this reaction for the given units?

A) $0.0417 [A][C]$ E) $4.17 [A][B]^2[C]$
B) $0.417 [A][B][C]$ F) $0.556 [A]^2[B]$
C) $0.0417 [A]^2[C]$ G) $0.278 [A]^2[C]$
D) $0.625 [B]^2[C]$ H) $0.208 [A][C]^2$

34. A solution of 0.10 M HF ($K_a = 7.2 \times 10^{-4}$) is adjusted to a pH of exactly 1.50. What is the solubility of CaF_2 ($K_{sp} = 3.9 \times 10^{-11}$) in this solution?

A) 16.7 mg/L D) 587 mg/L G) 72.0 mg/L
B) 305 ng/L E) 465 $\mu\text{g/L}$ H) 614 mg/L
C) 614 $\mu\text{g/L}$ F) 587 $\mu\text{g/L}$ I) 3.9 ng/L

35. The following is directly from a table of standard potentials (E°):



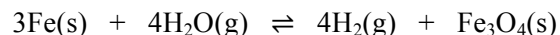
Calculate the value for K_{sp} for CdC_2O_4 .

A) 8.8×10^{-5} D) 2.3×10^{-18}
B) 9.4×10^{-3} E) 1.1×10^4
C) 5.9×10^{-32} F) 1.5×10^{-9}

36. *Ice vs Steam*: A mass ratio of 4:1 of ice to steam is put into an insulated container. The ice is introduced at -15°C and the steam at 115°C . What is the temperature in the container after thermal equilibrium is reached?

A) 20.6°C D) 68.2°C
B) 59.6°C E) 40.4°C
C) 35.3°C F) 13.9°C

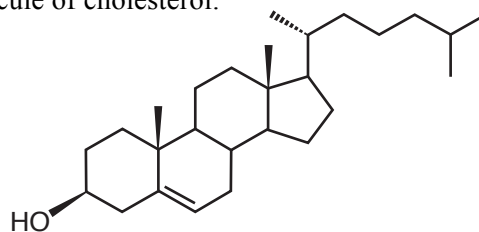
37. At 600 K, $K_p = 5.20$ for the following reaction:



30.74 g of H_2O and excess iron is added and sealed into a 12-liter reaction container. The container is then heated to 600 K and equilibrium is reached. What is the pressure of hydrogen gas at this temperature in this reaction mixture?

A) 2.79 atm D) 5.87 atm
B) 4.87 atm E) 3.85 atm
C) 5.37 atm F) 4.21 atm

38. The following is the abbreviated line structure for a molecule of cholesterol.



How many total hydrogen atoms are in this structure? Your total should include both the shown hydrogen atoms and any implied hydrogen atoms.

A) 52 C) 39 E) 45
B) 43 D) 46 F) 49

39. The reaction, $A \rightarrow 2B$, is found to be 2nd order in reactant A. The half-life for this reaction when A is initially 0.64 M, is 12.5 minutes. If the concentration of A is now started at 0.256 M with no B present. How long will it take for the product B to increase from 0 to 0.448 M?

A) > 10 hr C) 37.5 min E) 87.5 min
B) 3.65 hr D) 4.32 hr F) < 5 min

40. Identify the molecule that has the most polar bonds and then the molecule with the greatest dipole moment from this list:
- F₂ CS₂ CH₃I NH₃ PF₅
- A) F₂ most polar bonds ; NH₃ greatest dipole
 B) PF₅ most polar bonds and the greatest dipole
 C) PF₅ most polar bonds ; CH₃I greatest dipole
 D) NH₃ most polar bonds and the greatest dipole
 E) PF₅ most polar bonds ; NH₃ greatest dipole
43. According to Tyson, as voted in August 2006 by the general assembly of the International Astronomical Union (IAU) that overturned the Planet Definition Resolution proposal by the Planet Definition Committee - Pluto was officially a dwarf planet because it failed which new criterion?
- A) A planet must be round.
 B) A planet must dominate the mass of its orbital zone.
 C) A planet must have a moon.
 D) A planet must have rings.
 E) A planet must orbit the Sun.

Physics Questions (41 – 60)

On this contest $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors in the $+x$, $+y$ and $+z$ directions respectively, and unless otherwise indicated, you may neglect air resistance in every problem.

41. According to Tyson, which state declared on March 8, 2007 that Pluto was a planet within state borders?
- A) Arizona
 B) California
 C) Nevada
 D) New Mexico
 E) Texas
42. According to Tyson, the American Dialect Society “[upgraded] the status of the word Pluto to a verb” to mean which of the following:
- A) to act as a laxative
 B) to demote or devalue
 C) to inflate
 D) to move far away
 E) to throw away
44. According to Tyson, in July 2008 the Executive Committee of the IAU Committee on Small Body Nomenclature approved a proposal for what name to classify “all dwarf planets that one might find orbiting beyond Neptune”?
- A) asteroids
 B) icy dwarf planets
 C) main plane planets
 D) plutoid
 E) trans-Neptunian object
45. For a particle moving in one dimension along the x -axis in which the acceleration a is a function of the position x that is given by $a(x) = 3.0x$, where x is in meters and a is in meters/second². If the velocity is zero at $x = 1.0$ m, then what is the velocity at $x = 5.0$ m?
- A) 3.5 m/s
 B) 4.9 m/s
 C) 8.5 m/s
 D) 8.7 m/s
 E) 42 m/s

46. An archer stands on a boulder and fires an arrow from a height of 6.50 m above the level ground below the boulder. If the initial velocity of the arrow fired from the bow is 53.0 m/s at an angle of 20.0° above the horizontal, then what is the horizontal distance that the arrow travels?
- A) 92.1 m
B) 107 m
C) 190 m
D) 201 m
E) 220 m
47. A 2500 kg truck is traveling uphill on a road that makes an angle of 23.0° with the horizontal. The coefficient of static friction between the truck tires and the road is 0.920. If the truck is speeding up, then what is the magnitude of the maximum acceleration of the truck?
- A) 4.47 m/s^2
B) 5.50 m/s^2
C) 8.31 m/s^2
D) 12.1 m/s^2
E) 12.5 m/s^2
48. A 5.0 kg particle starts from rest at $x = 0.025 \text{ m}$ and moves along the x -axis under a force $F(x) = 8.0 + 2.0x - 4.0x^2$, where x is in meters and F is in newtons. What is the work done by the force as the particle moves from $x = 0.025 \text{ m}$ to $x = 2.75 \text{ m}$?
- A) 1.6 J
B) 1.8 J
C) 20 J
D) 42 J
E) 57 J
49. A pendulum consists of a string of length L with a mass m attached at one end. The other end of the string is freely attached to a support. The mass is held such that the string is horizontal before being released from rest. A peg is located a distance R up from the lowest point in the motion of the mass. If the string is to stay taut as the mass swings around the peg in a full circle, then what is the maximum value of R ? You may model the mass as a point mass and neglect the thickness of the peg.
- A) $(2/5)L$
B) $(1/2)L$
C) $(3/5)L$
D) $(4/5)L$
E) L
50. A 25.0 g ball hits a vertical wall. The ball is in contact with the wall for 1.25 ms before bouncing off with the same speed and a reversed horizontal component of the velocity. The ball moves with a speed of 9.50 m/s and makes an angle of 30.0° with the normal to the wall both before and after the collision. What is the magnitude of the average, net, horizontal force exerted on the ball by the wall?
- A) 0.00 N
B) 219 N
C) 238 N
D) 329 N
E) 411 N
51. A harmonic, transverse wave on a string has the following wave function:
 $y(x,t) = (4.25\text{mm}) \sin(27.3x + 252t)$, where x is in meters and t is in seconds. What is the wave speed?
- A) $1.08 \times 10^{-1} \text{ m/s}$
B) $1.16 \times 10^{-1} \text{ m/s}$
C) $1.07 \times 10^0 \text{ m/s}$
D) $9.23 \times 10^0 \text{ m/s}$
E) $6.88 \times 10^3 \text{ m/s}$

52. How much heat is released when 250 g of steam at 175°C changes to 250 g of ice at 0.00°C ? Given the following data about water: density = $1,000\text{ kg/m}^3$, specific heat of ice = $2.05\text{ kJ/(kg} \cdot \text{K)}$, specific heat of water = $4.18\text{ kJ/(kg} \cdot \text{K)}$, specific heat of steam = $2.02\text{ kJ/(kg} \cdot \text{K)}$, latent heat of fusion = 333.5 kJ/kg , and latent heat of vaporization = 2257 kJ/kg .
- A) 143 kJ
B) 224 kJ
C) 226 kJ
D) 457 kJ
E) 790 kJ
53. A nonconducting sphere with a radius of 9.0 cm has a uniform volume charge density of $+625\text{ nC/m}^3$. What is the magnitude of the electric field at a distance of 6.0 cm from the center of the sphere?
- A) $0.0 \times 10^0\text{ N/C}$
B) $1.3 \times 10^{-8}\text{ N/C}$
C) $2.1 \times 10^{-1}\text{ N/C}$
D) $1.7 \times 10^1\text{ N/C}$
E) $1.4 \times 10^3\text{ N/C}$
54. A $2.4\text{ }\mu\text{F}$ capacitor is charged to 60 V. After being fully charged the capacitor is disconnected from the voltage source and is connected to a completely discharged second capacitor without losing any charge. If the final voltage across the $2.4\text{ }\mu\text{F}$ capacitor is 20 V, then what is the capacitance of the second capacitor?
- A) $1.2\text{ }\mu\text{F}$
B) $2.4\text{ }\mu\text{F}$
C) $4.8\text{ }\mu\text{F}$
D) $7.2\text{ }\mu\text{F}$
E) $9.6\text{ }\mu\text{F}$
55. A particle has mass m , charge q , and moves perpendicular to a uniform magnetic field B in a circular orbit with radius R . What is the kinetic energy of the particle?
- A) qBR
B) $q^2B^2R^2$
C) $(qBR)/(2m)$
D) $(q^2B^2R^2)/(2m)$
E) $(mq^2B^2R^2)/2$
56. A long solenoid has 75 turns/cm and carries a 5.0 A current. The solenoid is filled with iron and has a magnetic field with a magnitude of 2.34 T at the center of the solenoid. What is the magnitude of the magnetization of the iron?
- A) $4.7 \times 10^{-2}\text{ A/m}$
B) $3.7 \times 10^4\text{ A/m}$
C) $8.8 \times 10^4\text{ A/m}$
D) $1.8 \times 10^6\text{ A/m}$
E) $1.9 \times 10^6\text{ A/m}$
57. A laser pulse has an average energy of 30 J and a beam radius of 1.5 mm. The pulse lasts for 12 ns. Determine the average energy density within the pulse.
- A) $1.2 \times 10^0\text{ J/m}^3$
B) $8.8 \times 10^2\text{ J/m}^3$
C) $1.8 \times 10^3\text{ J/m}^3$
D) $5.9 \times 10^5\text{ J/m}^3$
E) $1.2 \times 10^6\text{ J/m}^3$
58. In a double-slit experiment in which the slits are each $12.0\text{ }\mu\text{m}$ wide and have a center to center separation of $24.0\text{ }\mu\text{m}$ that is illuminated with 600 nm light and the interference pattern is viewed on a screen that lies 4.00 m away from the center of the slits. If a point P lies 70.0 cm away from the center of the pattern on the screen, then what is the ratio of the intensity at point P to the intensity at the center of the pattern?
- A) 2.60×10^{-4}
B) 7.43×10^{-3}
C) 8.30×10^{-3}
D) 2.26×10^{-1}
E) 8.95×10^{-1}
59. A telescope that has a circular aperture with a diameter of 250 inches is used to image a double star 8.0 light-years away. If light with a wavelength of 625 nm is used, then what is the minimum distance between the two stars so the image will be resolved? Assume ideal viewing conditions.
- A) $7.5 \times 10^7\text{ m}$
B) $1.6 \times 10^8\text{ m}$
C) $6.1 \times 10^9\text{ m}$
D) $9.1 \times 10^9\text{ m}$
E) $4.7 \times 10^{10}\text{ m}$

60. The probability of muon decay is $P(t) = 1 - e^{-t/\tau}$, where τ is the mean proper lifetime which is $2.12 \mu\text{s}$. If a muon is created 9000 m above the surface of the Earth, then what is the probability that a muon traveling at $0.980c$ will make it to the surface of the Earth?
- A) $5.36 \times 10^{-5}\%$
 - B) 5.65%
 - C) 50.0%
 - D) 94.3%
 - E) 100% (to 3 significant digits)

UIL HIGH SCHOOL SCIENCE CONTEST

Contestant Answer Sheet

Contestant # _____

9 10 11 12
CONTESTANT GRADE LEVEL

A 2A 3A 4A 5A
CONFERENCE

PLEASE WRITE ANSWERS WITH CAPITAL LETTERS

- | | | |
|-----------|-----------|-----------|
| 1. _____ | 21. _____ | 41. _____ |
| 2. _____ | 22. _____ | 42. _____ |
| 3. _____ | 23. _____ | 43. _____ |
| 4. _____ | 24. _____ | 44. _____ |
| 5. _____ | 25. _____ | 45. _____ |
| 6. _____ | 26. _____ | 46. _____ |
| 7. _____ | 27. _____ | 47. _____ |
| 8. _____ | 28. _____ | 48. _____ |
| 9. _____ | 29. _____ | 49. _____ |
| 10. _____ | 30. _____ | 50. _____ |
| 11. _____ | 31. _____ | 51. _____ |
| 12. _____ | 32. _____ | 52. _____ |
| 13. _____ | 33. _____ | 53. _____ |
| 14. _____ | 34. _____ | 54. _____ |
| 15. _____ | 35. _____ | 55. _____ |
| 16. _____ | 36. _____ | 56. _____ |
| 17. _____ | 37. _____ | 57. _____ |
| 18. _____ | 38. _____ | 58. _____ |
| 19. _____ | 39. _____ | 59. _____ |
| 20. _____ | 40. _____ | 60. _____ |

BIOLOGY SCORE

CHEMISTRY SCORE

PHYSICS SCORE

OVERALL SCORE:

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY**

STATE • 2015

- | | | |
|-------|-------|-------|
| 1. B | 21. E | 41. D |
| 2. D | 22. B | 42. B |
| 3. A | 23. D | 43. B |
| 4. C | 24. A | 44. D |
| 5. E | 25. C | 45. C |
| 6. D | 26. D | 46. D |
| 7. C | 27. C | 47. A |
| 8. D | 28. B | 48. A |
| 9. A | 29. A | 49. A |
| 10. E | 30. D | 50. D |
| 11. C | 31. E | 51. D |
| 12. A | 32. F | 52. E |
| 13. A | 33. G | 53. E |
| 14. D | 34. C | 54. C |
| 15. C | 35. A | 55. D |
| 16. D | 36. B | 56. D |
| 17. E | 37. F | 57. E |
| 18. E | 38. D | 58. B |
| 19. B | 39. B | 59. D |
| 20. E | 40. E | 60. B |

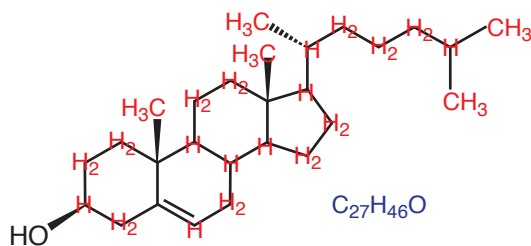
UIL HIGH SCHOOL SCIENCE CONTEST SOLUTIONS

CHEMISTRY

21. (E) $2\text{MnO}_4^- + 10\text{Cl}^- + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{Cl}_2 + 8\text{H}_2\text{O}$
 ratio Cl_2 to MnO_4^- is 5/2
 $80\text{mL} \times .05\text{M} = 4 \text{ mmol}$ (limiting, needs 20 mmol Cl^-)
 $320\text{mL} \times 0.075\text{M} = 24 \text{ mmol}$ (excess by 4 mmol)
 $722/760 = 0.95 \text{ atm}$
 $4 \text{ mmol} (5/2) = 10 \text{ mmol } \text{Cl}_2$ (0.01 mol Cl_2)
 $V = nRT/P = .01(.08206)(301.15)/.95 = 0.26013 \text{ L}$
 $V = 260 \text{ mL}$
22. (B) $\text{CS}_2 + 3 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{SO}_2$
 ratio SO_2 to CS_2 is 2/1
 $62.3/76.215 = 0.8181 \text{ mmol } \text{CS}_2$
 $0.8181 (2/1) = 1.636 \text{ mmol } \text{SO}_2$ (matches 100% yield)
 $86.2/64.07 = 1.3454 \text{ mmol actual yield}$
 $1.3454/1.636 \times 100\% = 82.2\% \text{ yield}$
23. (D) Break(+):
 $\text{C-O}, \text{C}\equiv\text{O} = 1072$
 Make(-): $\text{C-C} = 346, \text{C=O} = 799, \text{C-O}$
 C-O's cancel
 $+1072 -346 -799 = -73$
24. (A) Use Arrhenius equation:
 $\ln(760/100) = 34000/8.314(1/T - 1/353.15)$
 $2.028 = 4089.5(1/T - 0.002832)$
 $1/T = 2.028/4089.5 + .002832 = 0.0033276$
 $T = 300.5 \text{ K} = 27.4^\circ\text{C}$
25. (C) Flip and double the first. Leave second as is. ADD
 $4 \text{HCl(g)} \rightarrow 2\text{H}_2\text{(g)} + 2 \text{Cl}_2\text{(g)} \quad +369.2 \text{ kJ}$
 $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2 \text{H}_2\text{O(g)} \quad -483.6 \text{ kJ}$
 giving -114.4 kJ
26. (D) $m_{\text{actual}} = \Delta T_f/K_f = 0.614/1.86 = 0.33 \text{ m}$
 $0.33/0.30 = 1.10 \rightarrow 10\% \text{ ionized}$
27. (C) The X drops from .80 to .45 on the plot which is a decrease of .35 M. The Y will increase from zero to double that which is .70 and matches plot C.
28. (B) $\text{pKb} = -\log(1.2\text{e-}6) = 5.9208$ tris is B
 $50(.15) = 7.5 \text{ mmol H}^+ \quad 200(.17) = 34.0 \text{ mmol B}$
 $34 - 7.5 = 26.5 \text{ mmol B left and } 7.5 \text{ mmol BH}^+ \text{ made}$
 $\text{pKa} = 14 - \text{pKb} = 14 - 5.9208 = 8.0792$
 $\text{pH} = \text{pKa} + \log(\text{B/BH}^+) = 8.0792 + \log(26.5/7.5)$
 $\text{pH} = 8.0792 + 0.5482 = 8.6274$
29. (A) The standard potential is zero (same $\frac{1}{2}$ reaction on each side). Left side is anode and therefore the .88 M cation is a product of the reaction. Use Nernst equation:
 $E = 0 - 0.0257/2 \ln(.88/.00011) = -0.115 \text{ V}$
30. (D) The pK_a for indicator is 4.49 which is where it would be green (50/50 mix of yellow and blue). The acid pH is calculated: $[\text{H}^+] = \sqrt{2.3 \times 10^{-9} \cdot 0.051} = 1.08 \times 10^{-5}$ which gives a pH of 4.97 which puts the indicator in a ratio of 3:1 of blue:yellow. That will give a color of blue-green.

UIL HIGH SCHOOL SCIENCE CONTEST SOLUTIONS

31. (E) The Rydberg equation: $\Delta E = R(1/n^2 - 1/n'^2)$
 $\Delta E = 2.178 \times 10^{-18}(1/2^2 - 1/6^2) = 2.178 \times 10^{-18}(0.2222)$
 $\Delta E = 4.84 \times 10^{-19} \text{ J} \quad \lambda = hc/\Delta E$
 $\lambda = 6.626 \times 10^{-34}(3 \times 10^8)/4.84 \times 10^{-19}$
 $= 4.11 \times 10^{-7} \text{ m} = 411 \text{ nm}$
32. (F) Quantum number rules:
 $n = 1, 2, 3, 4, \dots$
 $\ell = 0, 1, 2, 3, \dots (n-1)$ corresponding to letters
 $= s, p, d, f, g, h, i, \dots$
 You cannot have $n = 3$ paired with $\ell = 3 (f)$
33. (G) Comparing 1 and 2 we know B is zero order.
 Comparing 1 and 3 we know C is first order.
 Comparing 1 and 4 we know A is 2nd order.
 Solve for $k = \text{rate}/[A]^2[C] = 0.00125/(.15^2 \cdot .2)$
 $k = 0.278$
34. (C) $K_a = [H^+][x]/.1-x$ where x is $[F^-]$
 $[F^-] = 7.2 \times 10^{-4}(.1)/(10^{-1.5} + 7.2 \times 10^{-4}) = 0.0022262 \text{ M}$
 $[Ca^{2+}] = 3.9 \times 10^{-11}/(0.0022262)^2 = 7.870 \times 10^{-6} \text{ M}$
 $7.870 \times 10^{-6} (78.08 \text{ g/mol}) = 6.145 \times 10^{-4} \text{ g/L}$
 $\times 1000 = 0.614 \text{ mg/L} = 614 \mu\text{g/L}$
35. (A) $E^\circ = -.522 - (-.402) = -.120 \text{ V}$
 $RT \ln K = nFE^\circ \quad K = \exp(nFE^\circ/RT)$
 $K = \exp\{2(96485)(-.120)/(8.314 \cdot 298.15)\}$
 $K = \exp\{-23156.4/2478.8\} = 8.769 \times 10^{-5}$
36. (B) Easiest to use 4g ice and 1g steam
 Take both to 100°C water as a point of reference
 ice: $15(2.09) + 334 + 418.4 \times 4 = 3135 \text{ J}$
 steam: $15(2.03) + 2260 = 2290.45 \text{ J}$
 difference is : 844.55 J on the cold side of 100°C
 $844.55/(4.184 \cdot 5\text{g}) = 40.4^\circ\text{C}$ down from 100°C
 which gives 59.6°C
37. (F) $30.74/18.016 = 1.706 \text{ mol H}_2\text{O}$
 $P = nRT/V = 1.706(0.08206)(600)/12 = 7.00 \text{ atm}$
 $K_p = P_{H_2}^4/P_{H_2O}^4 = [x/(7-x)]^4$
 $(5.2)^{1/4} = x/(7-x) \quad 1.51(7) = 2.51x \quad x = 4.21 = P_{H_2}$
38. (D) Structure with implied C's and H's. H = 46



UIL HIGH SCHOOL SCIENCE CONTEST SOLUTIONS

39. (B) 2^{nd} order $k = 1/[A]_0 t_{1/2} = 1/.64(12.5) = 0.125$
 product at .448 M means .224 M of reactant gone
 $.256 - .224 = 0.032$ for A at that time.
 $(1/.032 - 1/.256)/.125 = 218.75 \text{ min} = 3.65 \text{ hr}$
40. (A) Periodic table trends and molecular geometry.
 The P-F bond is the most polar bonds of those listed (biggest EN difference). Symmetry zeros the dipole for PF_5 , CS_2 , and F_2 . The NH_3 will have a bigger dipole moment than CH_3I .

PHYSICS

41. (D) "New Mexico is longtime home of Pluto discoverer Clyde Tombaugh... The state legislature felt that the IAU had unjustifiably dissed Pluto and, by association, their great state. So on March 8, 2007, their 48th legislature, in a bill introduced by Representative Joni Marie Gutierrez, passed a Joint Memorial declaring Pluto a planet within state borders..."
42. (B) "While astrophysicists were downgrading the cosmic object we call Pluto, the American Dialectic Society, which is more than a century old, was upgrading the status of the word Pluto to a verb, making it their 17th annual "Word of the Year" for 2006:
to pluto / to be plutoed: to demote or devalue someone or something ..."
43. (B) "Plutophiles had about a week to rejoice before learning the sad news that Pluto fails a new criterion – that a legitimate planet must also dominate the mass of its orbital zone."
44. (D) "At a July 2008 meeting in Oslo, Norway, the Executive Committee of the IAU approved a proposal from the IAU Committee on Small Body Nomenclature for the name *plutoid* to classify all dwarf planets that one might find orbiting beyond Neptune."
45. (C) By $a = dv/dt = (dv/dx)(dx/dt) = v(dv/dx) \Rightarrow \int v dv = \int a dx = \int 3x dx \Rightarrow [v^2/2]_0^v = [(3/2)x^2]_1^5$
 $\Rightarrow v = [3(5)^2 - 3(1)^2]^{1/2} = 8.5 \text{ m/s}$
46. (D) By $y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \Rightarrow 0 = 6.5 + 53(\sin 20^\circ)t - 4.9t^2 \Rightarrow t = 4.029 \text{ s}$
 $\& x = x_0 + v_{0x}t = 0 + 53(\cos 20^\circ) 4.029 = 201 \text{ m}$
47. (A) From N2L & the FBD of the situation: $\Sigma \mathbf{F}_\perp = +F_N - F_{G_\perp} = 0 \Rightarrow F_N = mg \cos 23^\circ$
 $\& \Sigma \mathbf{F}_\parallel = +F_{fs} - F_{G_\parallel} = ma \Rightarrow \mu_s mg \cos 23^\circ - mg \sin 23^\circ = ma$
 $\Rightarrow a = (9.8)[(0.92) \cos 23^\circ - \sin 23^\circ] = 4.47 \text{ m/s}^2$
48. (A) By $W = \int F dx = \int (8 + 2x - 4x^2) dx = [8x + x^2 - (4/3)x^3]_{0.025}^{2.75} = 1.65 \text{ J}$
49. (A) From N2L and the FBD at the top of the circle as it goes around the peg: $\Sigma \mathbf{F}_c = +F_T + F_G = ma_c$, for minimum speed: $0 + mg = mv_{\min}^2/R \Rightarrow v_{\min} = (gR)^{1/2}$ & by conservation of mechanical energy with the zero of gravitational potential energy at the top of the circle:
 $E_0 = E_{\text{top}} \Rightarrow 0 + mg(L - 2R) = \frac{1}{2}m(gR) + 0 \Rightarrow R = (2/5)L$
50. (D) By $\Sigma \mathbf{F}_{\text{avg},x} = \Delta \mathbf{p}_x / \Delta t = 2(0.025)(9.5 \cos 30^\circ) / (0.00125) = 329 \text{ N}$
51. (D) $v = \omega/k = 252/27.3 = 9.23 \text{ m/s}$

UIL HIGH SCHOOL SCIENCE CONTEST SOLUTIONS

52. (E) $Q = mc_{\text{steam}}\Delta T_{\text{steam}} - mL_V + mc_{\text{water}}\Delta T_{\text{water}} - mL_F$
 $= (0.25)(2.02)(100-175) - (0.25)(2257) + (0.25)(4.18)(0-100) - (0.25)(333.5) = -790 \text{ kJ}$,
 so 790 kJ of heat is released
53. (E) By Gauss' law: $\oint \mathbf{E} \cdot d\mathbf{a} = Q_{\text{enc}}/\epsilon_0 \Rightarrow E(4\pi r^2) = \rho[(4/3)\pi r^3]/\epsilon_0$
 $\Rightarrow E = [\rho r]/[3\epsilon_0] = [(625\text{E-}9)(0.06)]/[3(8.85\text{E-}12)] = 1.4\text{E+}3 \text{ N/C}$
54. (C) $Q_0 = (2.4\text{E-}6)(60) = 144\mu\text{C}$ & $Q_1 = (2.4\text{E-}6)(20) = 48\mu\text{C}$, with $Q_0 = Q_1 + Q_2 \Rightarrow Q_2 = 96\mu\text{C}$,
 thus $C = (96\text{E-}6)/(20) = 4.8\mu\text{F}$
55. (D) By $\Sigma \mathbf{F}_c = +F_B = ma_c \Rightarrow |q|vB \sin 90^\circ = mv^2/R \Rightarrow mv = |q|BR \Rightarrow K = \frac{1}{2}mv^2 = (q^2 B^2 R^2)/(2m)$
56. (D) $B_{\text{app}} = \mu_0 nI = (4\pi\text{E-}7)(7500)(5) = 4.71\text{E-}2 \text{ T}$, with $B = B_{\text{app}} + \mu_0 M$
 $\Rightarrow M = (2.34 - 4.71\text{E-}2)/(4\pi\text{E-}7) = 1.8\text{E+}6 \text{ A/m}$
57. (E) By $I = U_{\text{avg}}/(A \cdot t) = u_{\text{avg}}c \Rightarrow u_{\text{avg}} = U_{\text{avg}}/[A \cdot t \cdot c] = 30/[\pi(1.5\text{E-}3)^2(12\text{E-}9)(3\text{E+}8)] = 1.2\text{E+}6 \text{ J/m}^3$
58. (B) With calculator in radian mode: $\theta = \tan^{-1}(y/L) = \tan^{-1}(0.7/4) = 0.1732 \text{ rad}$,
 then $\alpha = [\pi a/\lambda] \sin \theta = [\pi(12\text{E-}6)/(600\text{E-}9)] \sin(0.1732 \text{ rad}) = 10.83 \text{ rad}$
 & $\beta = [\pi d/\lambda] \sin \theta = [\pi(24\text{E-}6)/(600\text{E-}9)] \sin(0.1732 \text{ rad}) = 21.66 \text{ rad}$
 $\Rightarrow I_p/I_m = [(\sin \alpha)/\alpha]^2 [\cos \beta]^2 = [(\sin(10.83 \text{ rad}))/ (10.83 \text{ rad})]^2 [\cos(21.66 \text{ rad})]^2 = 7.43\text{E-}3$
59. (D) Using the small angle approximation and the Rayleigh criterion for resolution:
 $s = R\theta_{\text{min}} = R(1.22\lambda/D) = [(8)(3\text{E+}8)(1.556\text{E+}7)]\{(1.22)(625\text{E-}9)/[(250)(2.54)(1/100)]\} = 9.1\text{E+}9 \text{ m}$
60. (B) With $\gamma = [1 - (v/c)^2]^{-1/2} = [1 - (0.98)^2]^{-1/2} = 5.025$,
 then $t = L_0/v = L/[\gamma v] = (9000)/[(5.025)(0.98)(3\text{E+}8)] = 6.092\text{E-}6 \text{ s}$
 & (prob. of making it) = $1 - (\text{prob. of decay}) = e^{-(t/\tau)} = e^{-(6.092\text{E-}6/2.12\text{E-}6)} = 0.0565 \Rightarrow 5.65\%$